ROB-GY 6103
Advanced Mechatronics

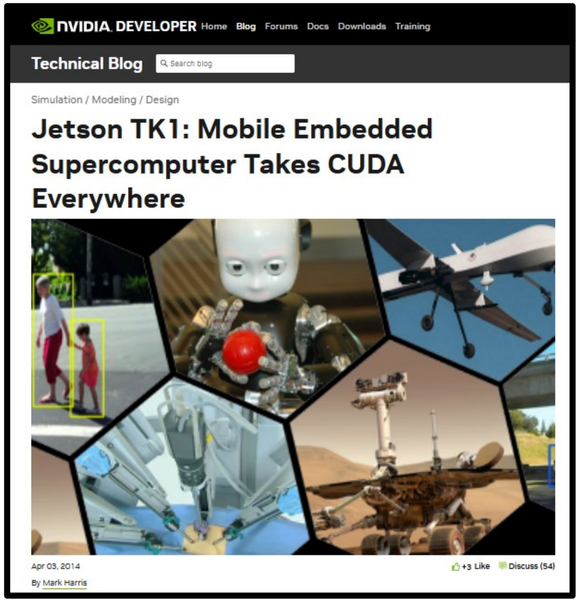
Lecture 8

Vikram Kapila

Mechanical and Aerospace Engineering

NYU Tandon School of Engineering

nVIDIA GPU-based Developer Kits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By Kevin Krewell, Former Contributor. Making technology comprehensible and relevant.
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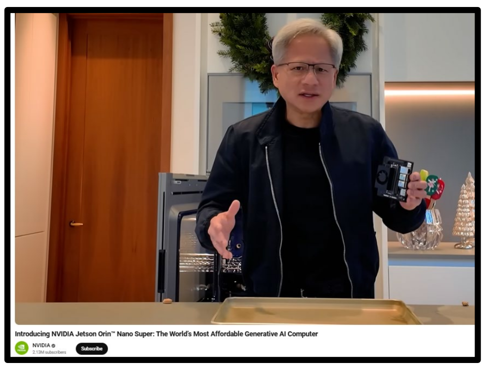

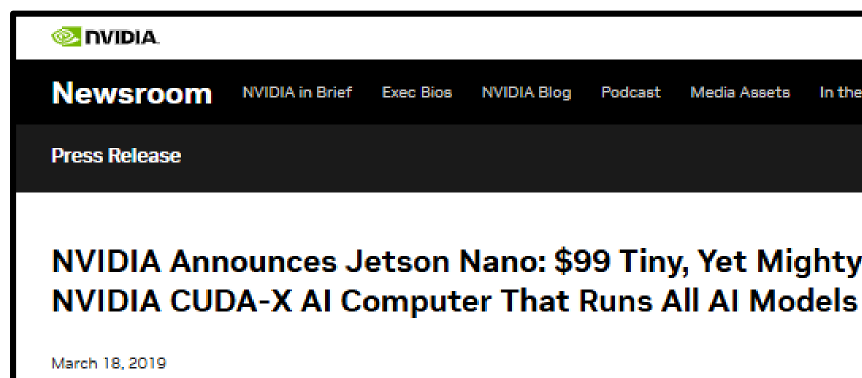
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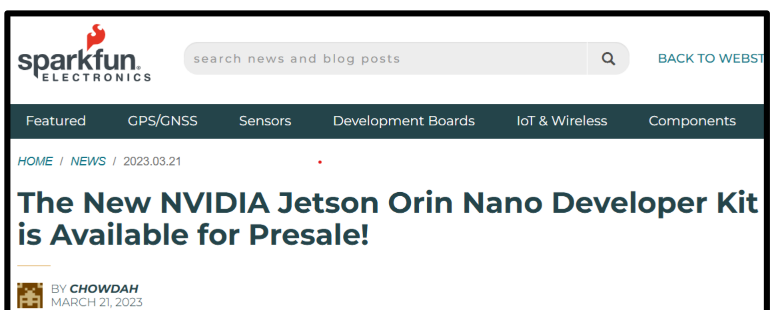
NVIDIA Announces Jetson Nano: $99 Tiny, Yet Mighty NVIDIA CUDA-X AI Computer That Runs All AI Models
March 18, 2019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The New NVIDIA Jetson Orin Nano Developer Kit is Available for Presale!
BY CHOWDAH
MARCH 21, 2023

Jetson Orin Nano Super Developer Kit

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NVIDIA Jetson Orin Nano Developer Kit Gets a "Super" Boost

Dec 17, 2024

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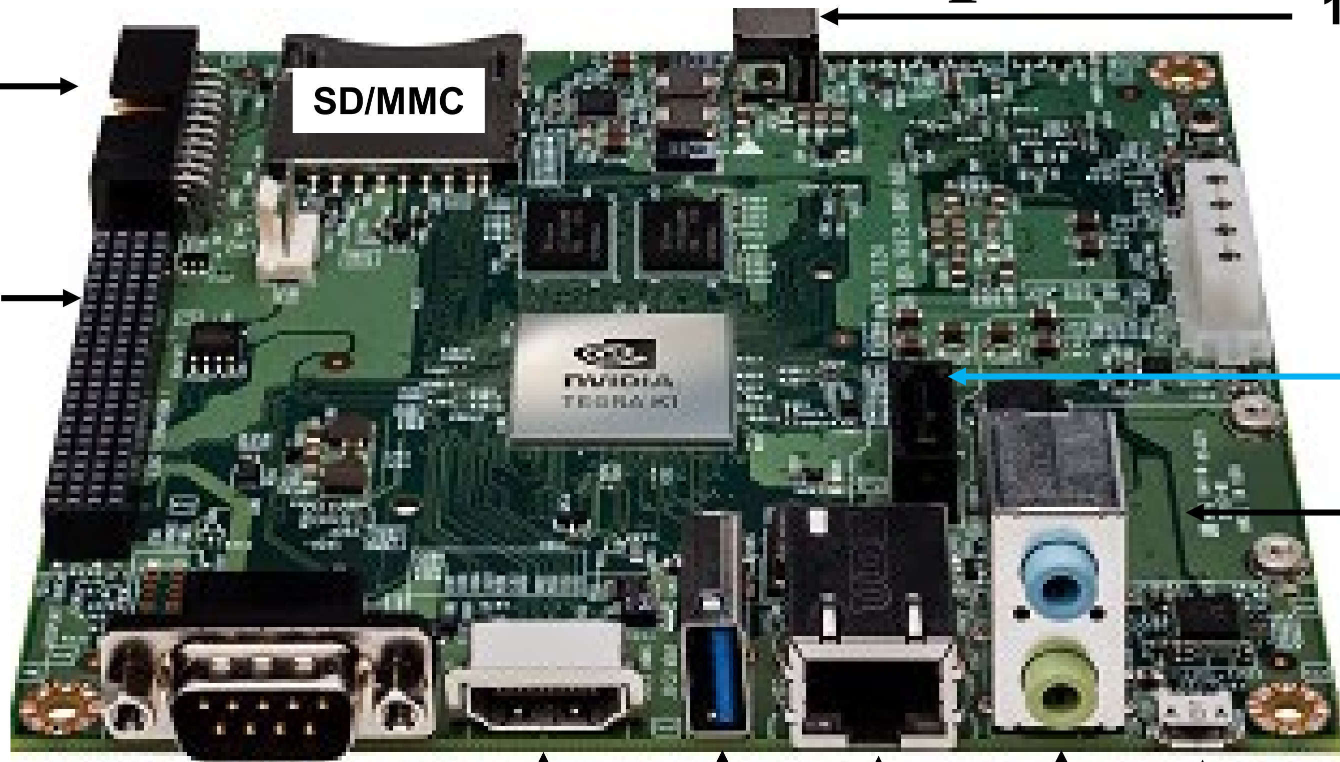
By Suhas Hariharapura Sheshadri, Leela Subramaniam Karumbunathan and Dustin Franklin

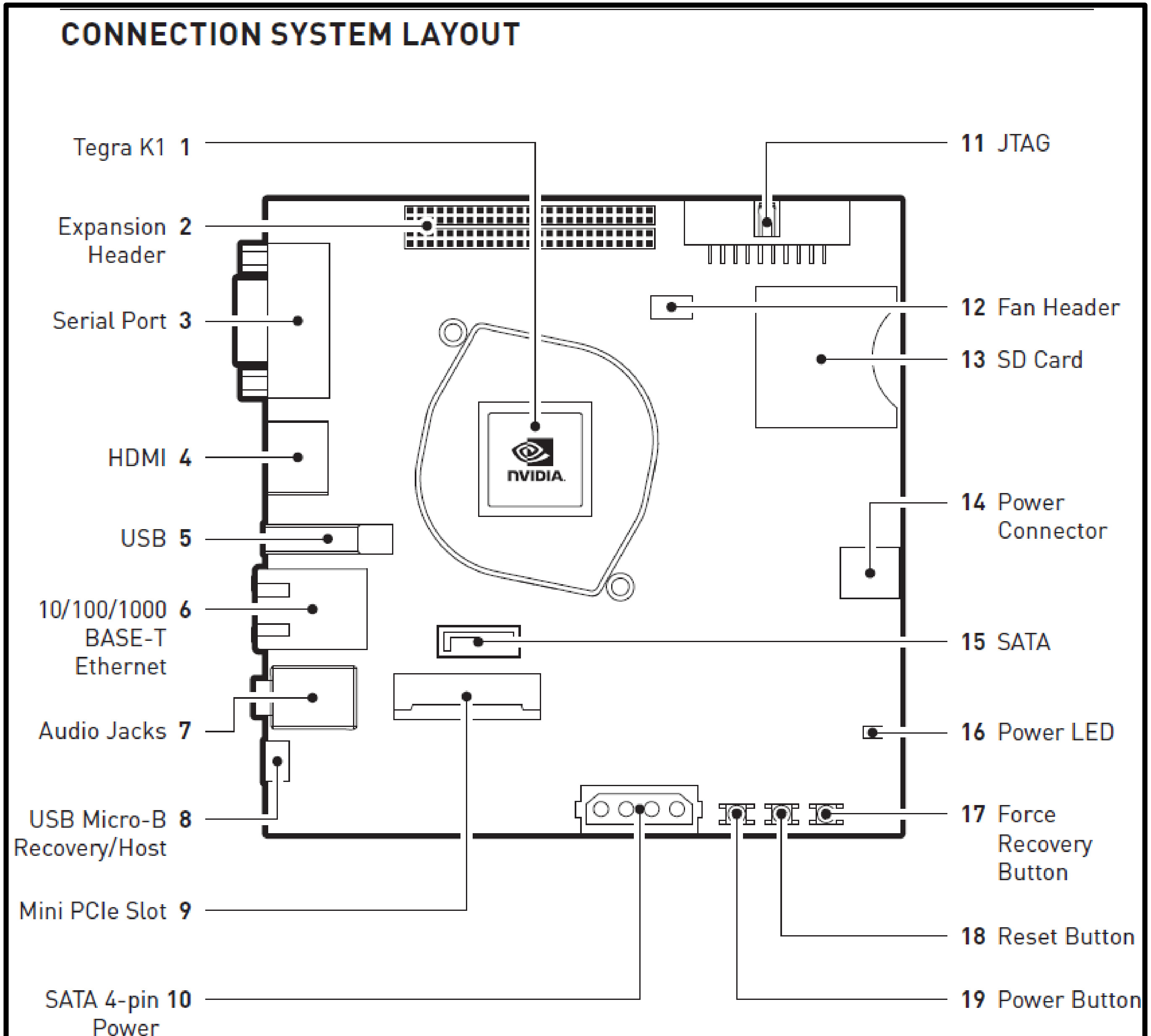
Jetson TK1 Developer Kit



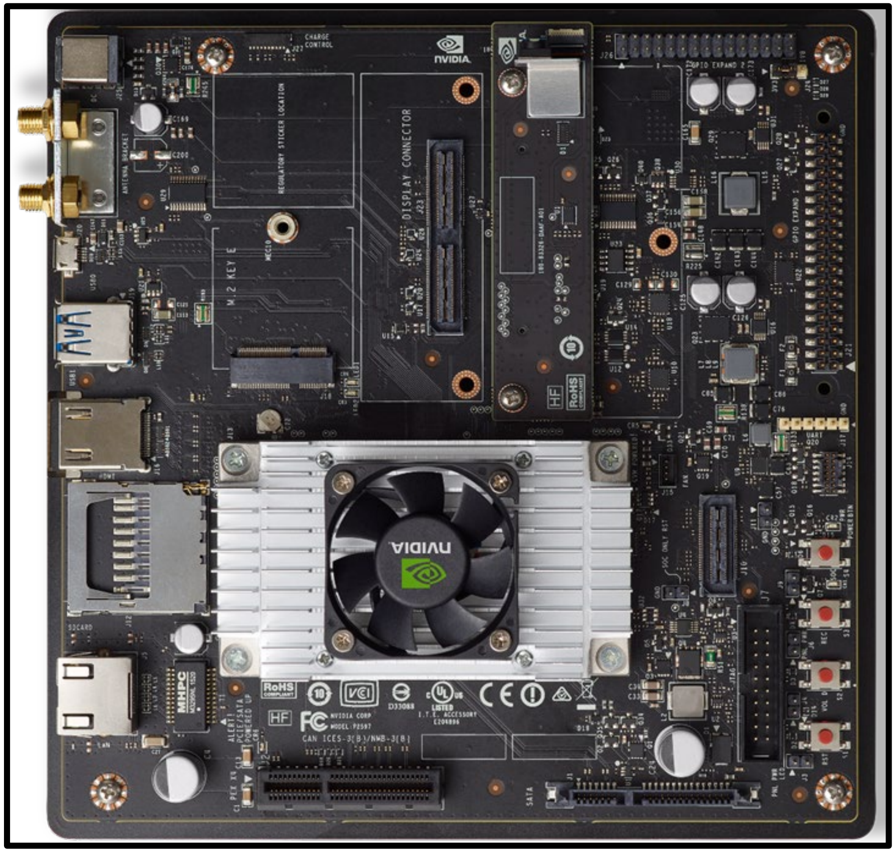
- CPU: Quad-core Arm Cortex-A15 @ 2.32GHz
- GPU: Kepler architecture , 192 CUDA cores @ 950MHz
- TFLOPS: 0.327 (FP32)
- Memory: 2GB
- Power: 10W
- Size: 127mm × 127mm × 26mm

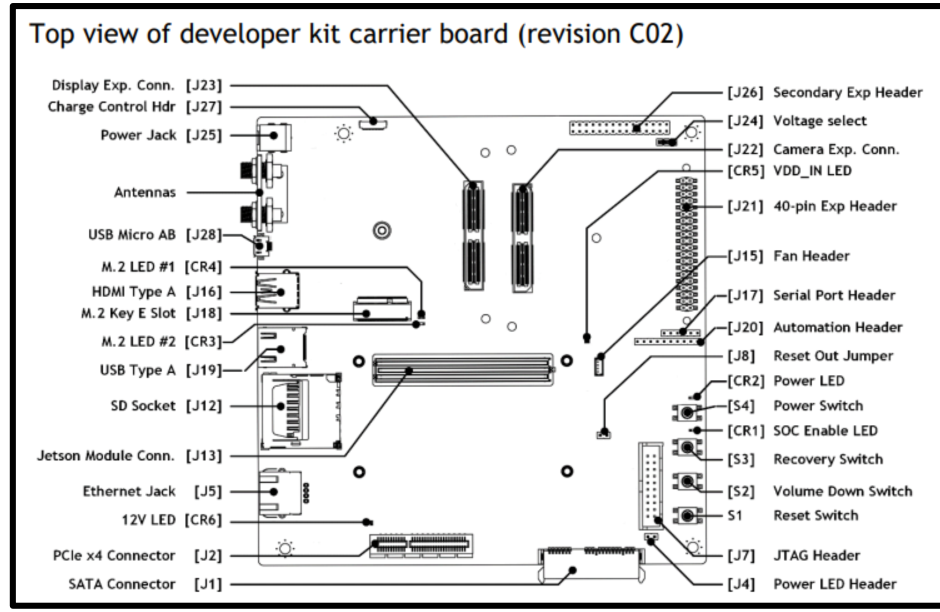
Released: April 2014
Reached End of Life (EOL)

Jetson TK1 Developer Kit



Jetson TX2 Developer Kit



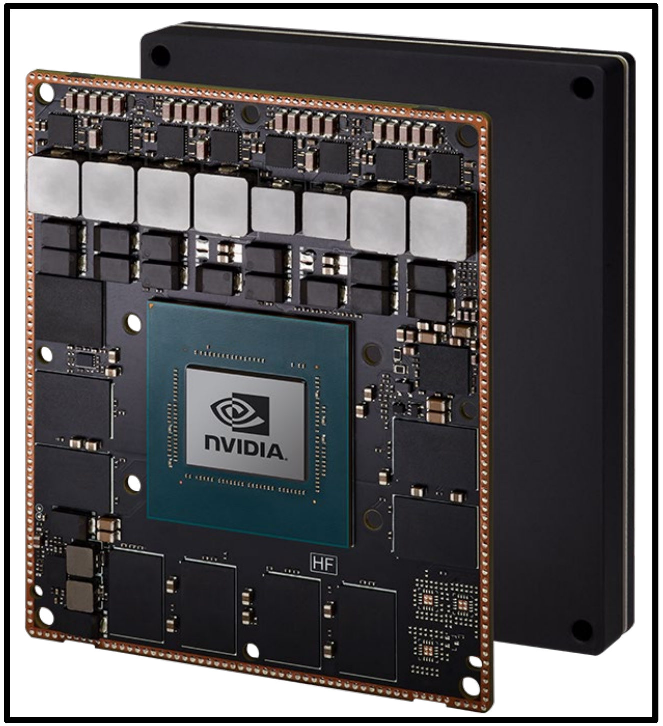


Top view of developer kit carrier board (revision C02)

- **CPU: Quad-core Arm Cortex-A57 @ 2GHz; Dual-Core NVIDIA Denver 2 @ 2GHz 64-Bit**
- **GPU: Pascal architecture, 256 CUDA cores, 1.3GHz**
- **TFLOPS: 1.3 (FP16)**
- **Memory: 8GB 128-bit RAM**
- **Power: 7.5W | 15W**
- **Size: 170mm × 170mm × 50mm**

Released: March 2017
Reached End of Life (EOL)

NVIDIA AGX XAVIER Developer Kit

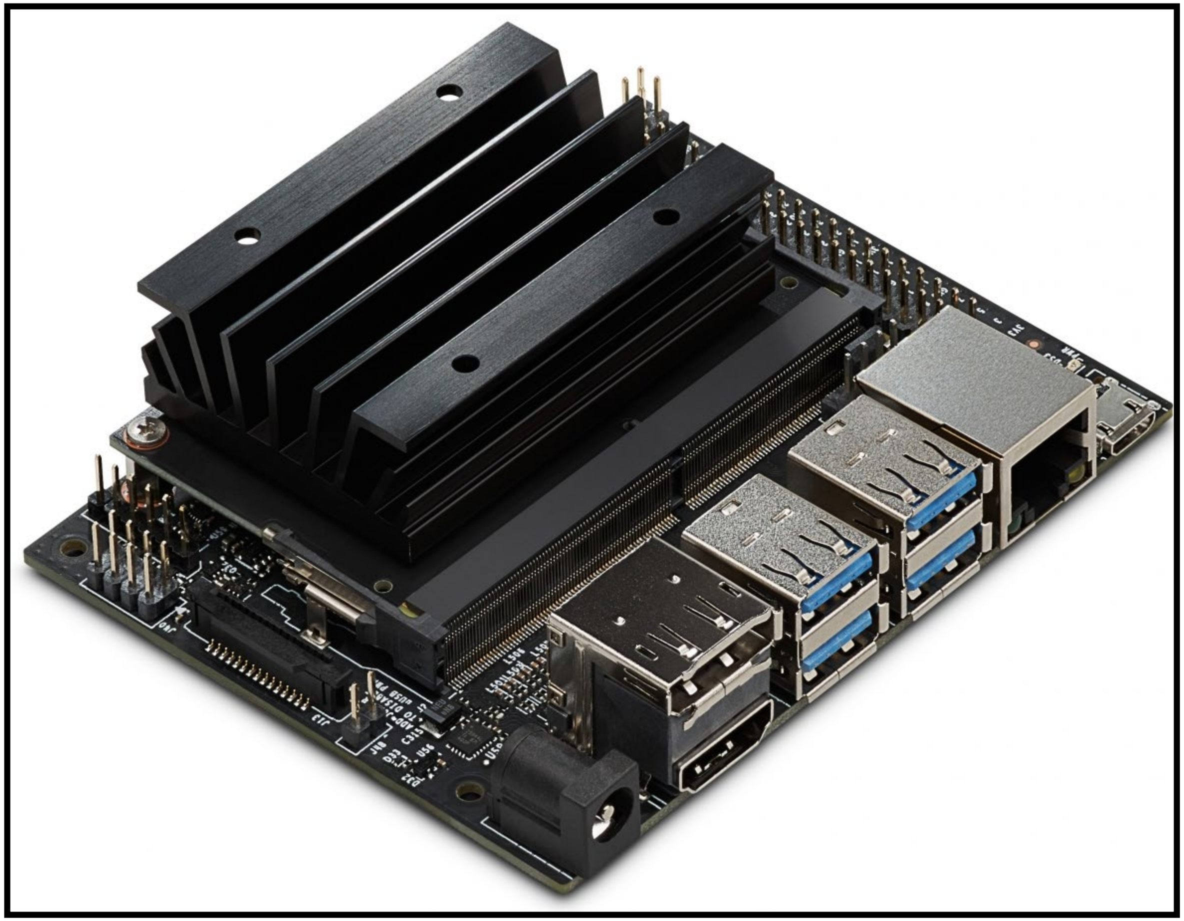


AGX XAVIER Module

- CPU: 8-Core, ARMv8.2 Carmel CPU
- GPU: Volta architecture, 512 CUDA cores, 64 Tensor Cores, 1.37GHz
- TFLOPS: 5.5-11 (FP16) TFLOPS | 20-32 (INT8) TOPS
- Memory: 32GB 256-bit RAM
- Power: 10W | 15W | 30W
- Size: 105mm × 105mm × 65mm

Released: August 2018
Reached End of Life (EOL)

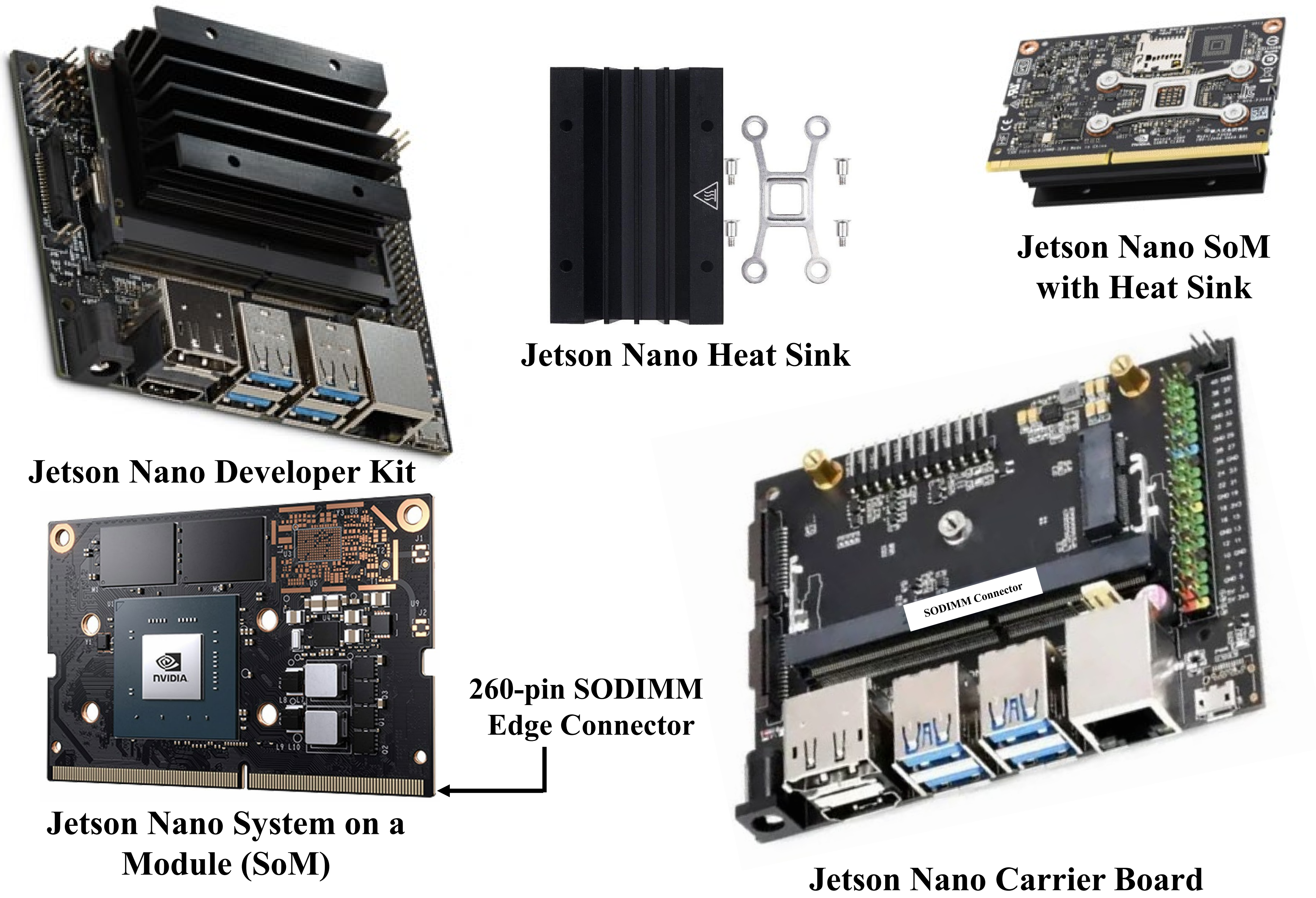
Jetson Nano Developer Kit



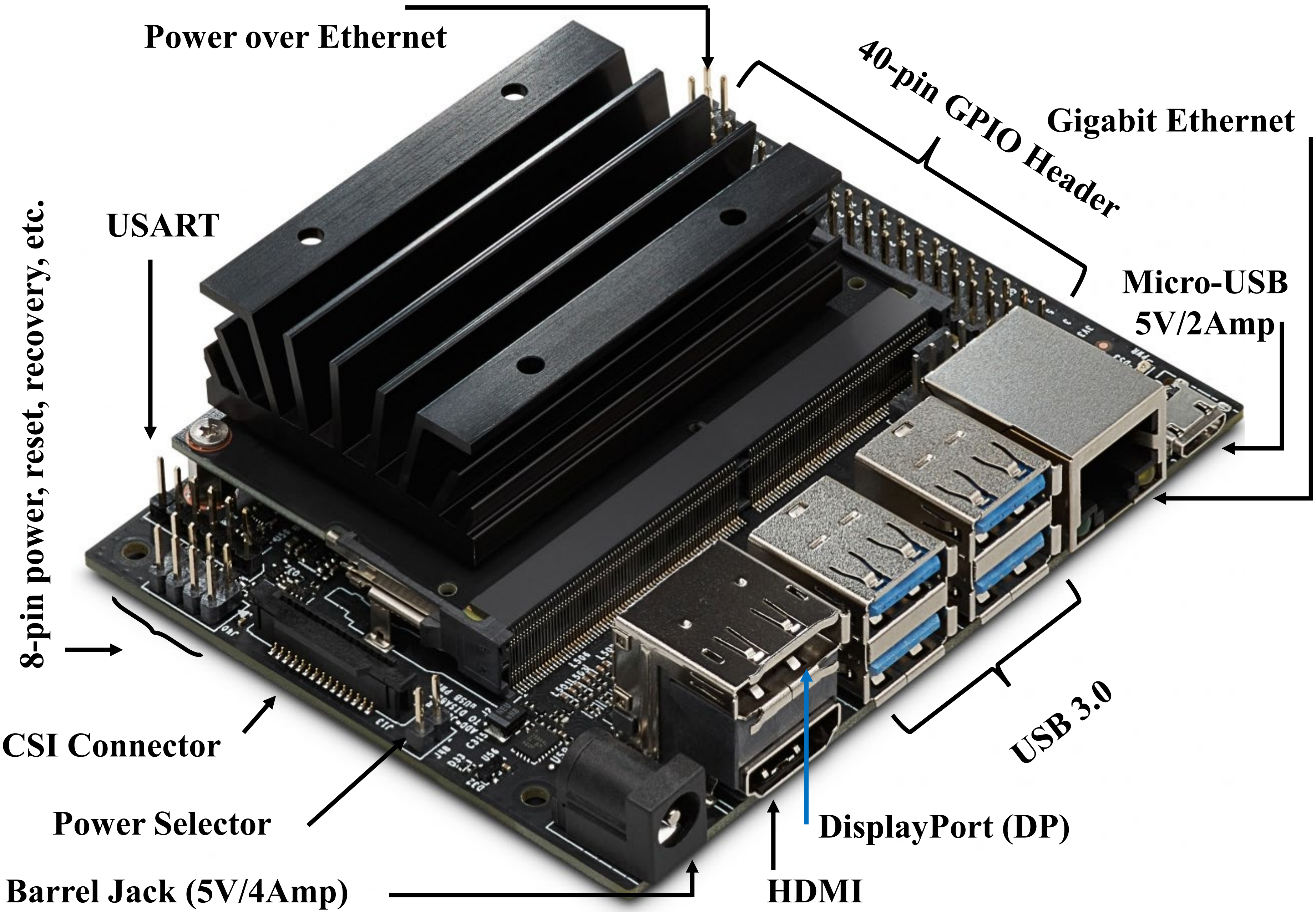
- **CPU: Quad-core, ARM A57, 1.43 GHz**
- **GPU: Maxwell architecture, 128 CUDA cores, 921MHz**
- **TFLOPS: 0.5 (FP16) [472 Giga FLOPS = 0.47TFLOPS]**
- **Memory: 4GB 64-bit**
- **Power: 5W | 10W**
- **Size: 100mm × 80mm × 29mm**

Released: March 2019
Reached End of Life (EOL)

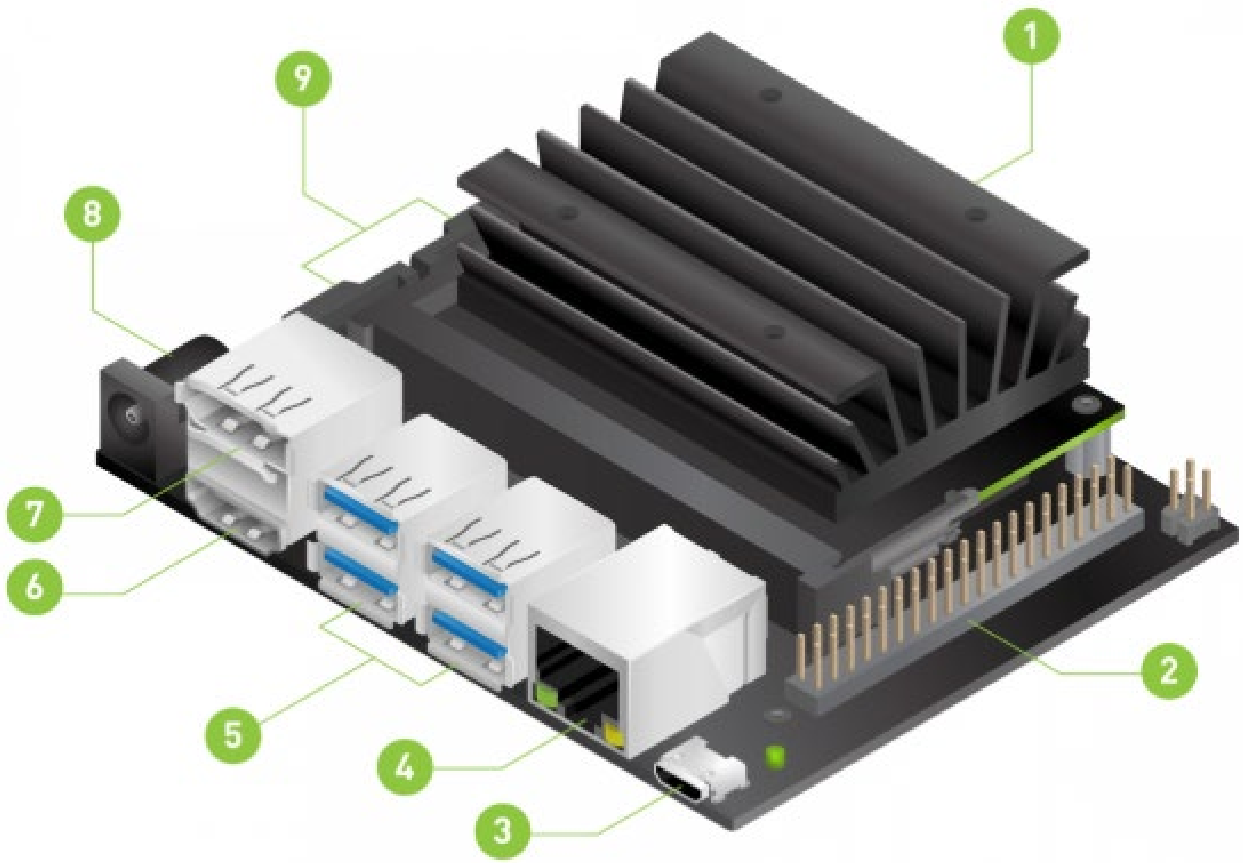
Jetson Nano Developer Kit



Jetson Nano Kit Anatomy—I

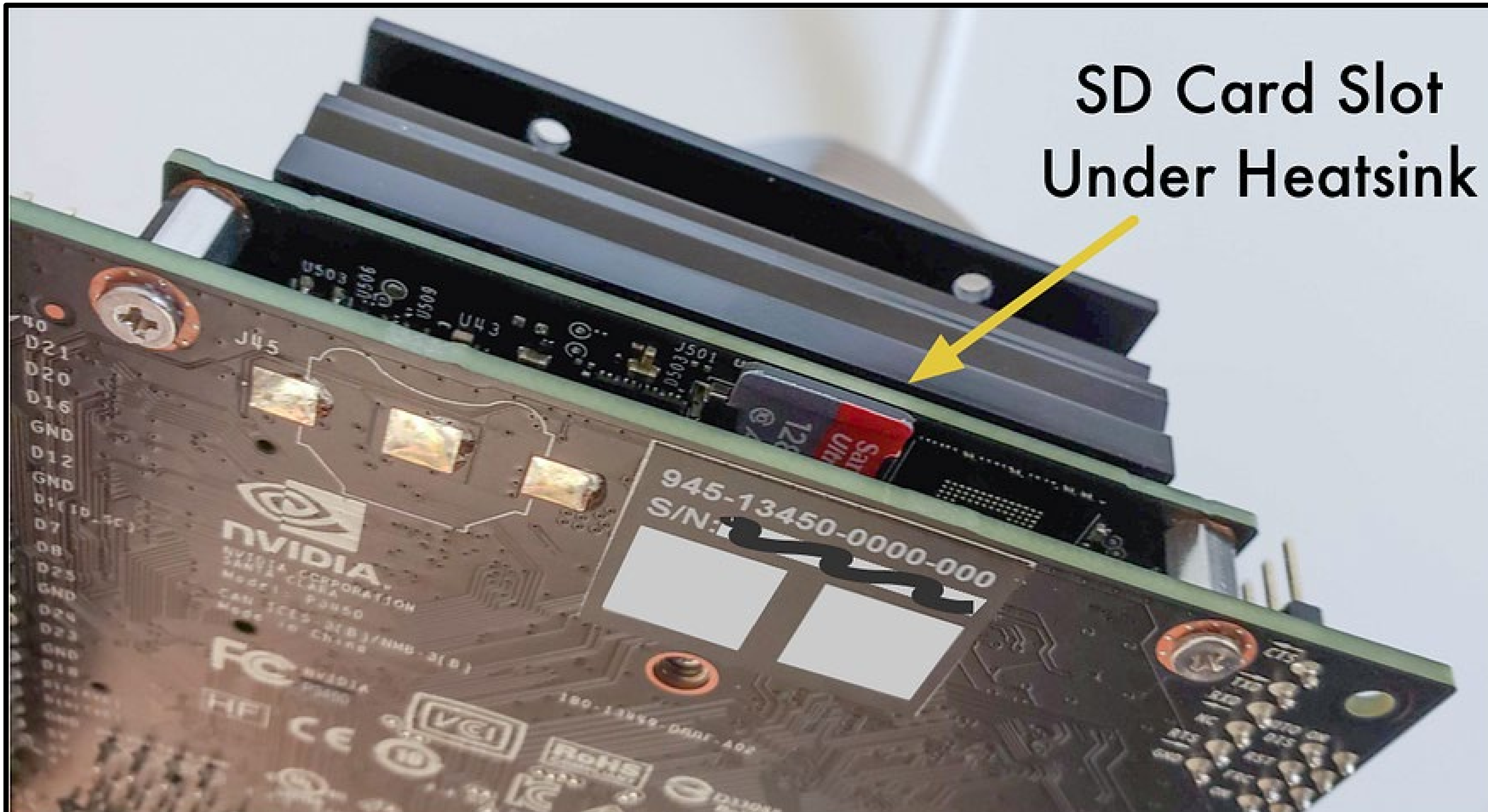


Jetson Nano Kit Anatomy—II



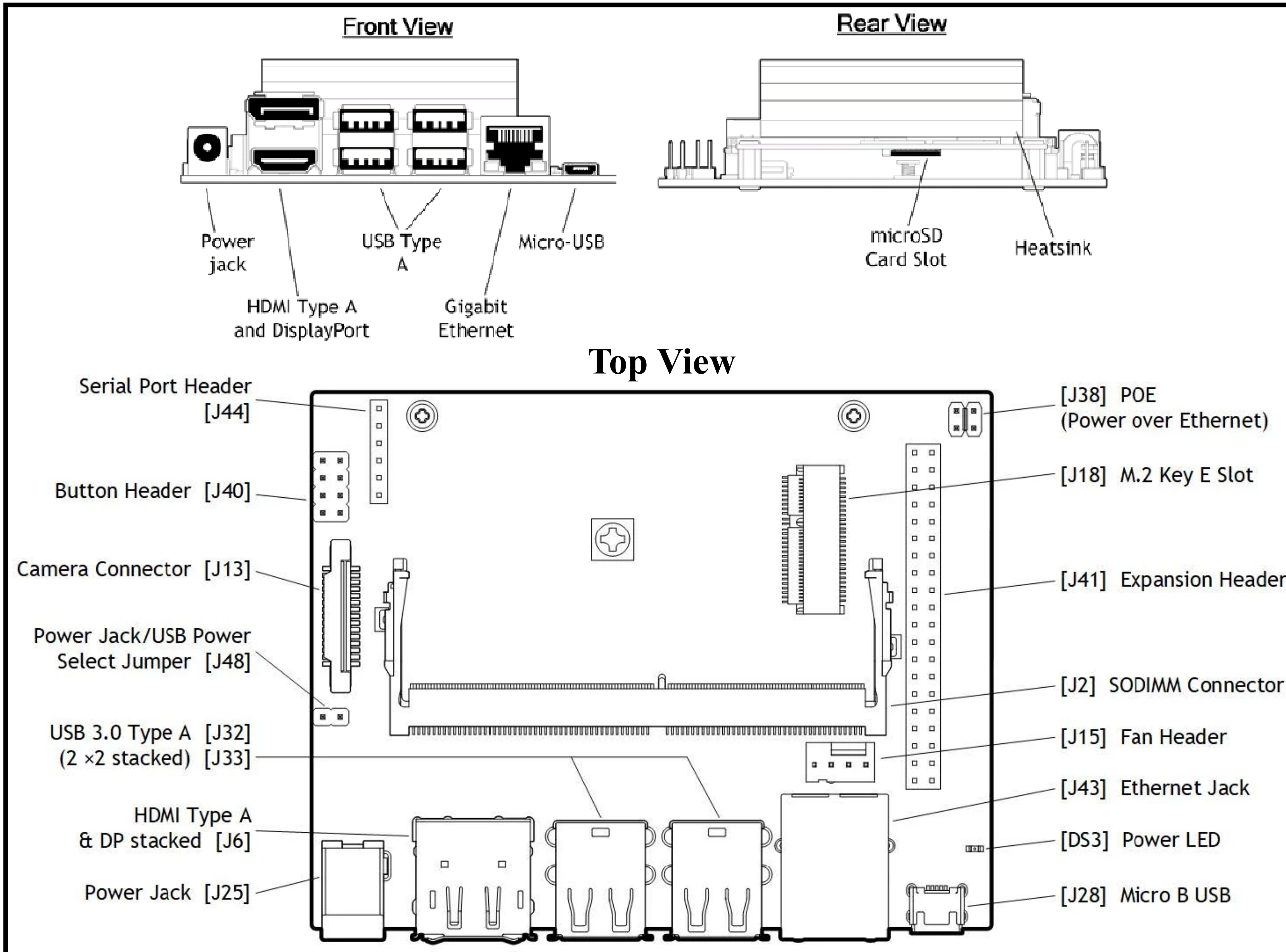
1. microSD card slot for main storage
2. 40-pin expansion header
3. Micro-USB port for 5V power input, or for Device Mode
4. Gigabit Ethernet port
5. USB 3.0 ports (x4)
6. HDMI output port
7. DisplayPort connector
8. DC Barrel jack for 5V power input
9. MIPI CSI-2 camera connectors

Jetson Nano Kit Anatomy—III

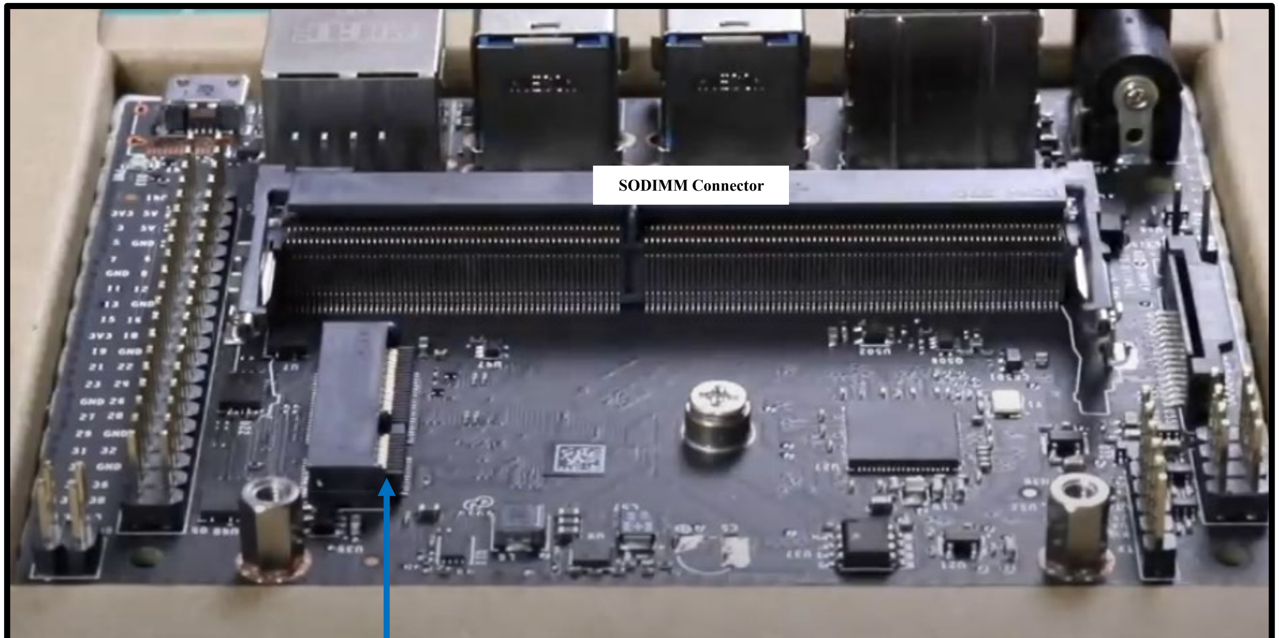


Jetson Nano Kit Anatomy—IV

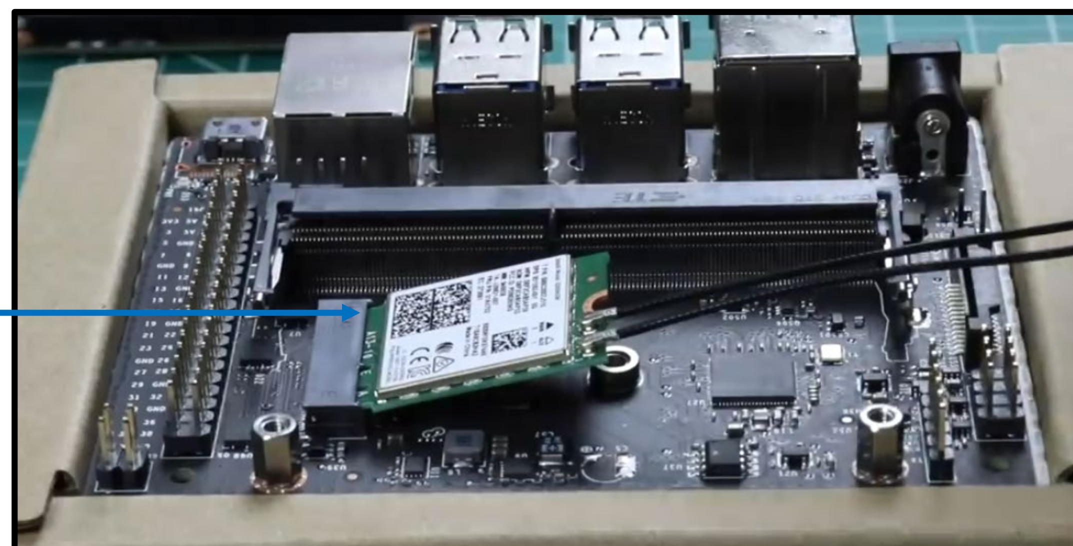
Jetson Nano Kit Anatomy—V



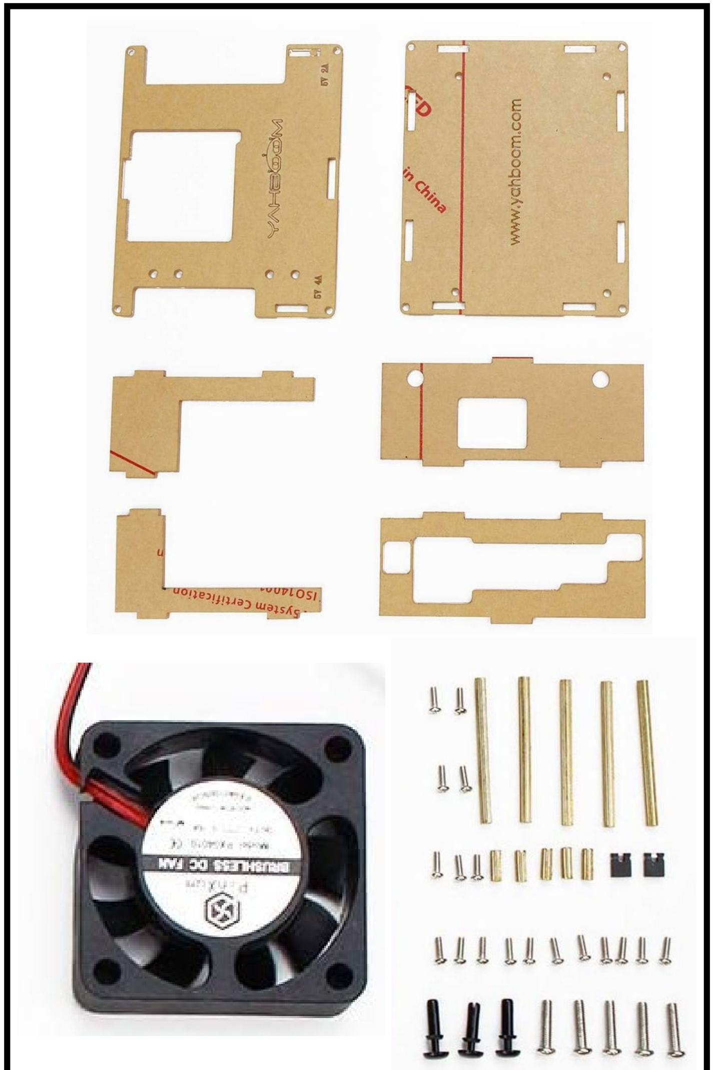
Jetson Nano Carrier Board

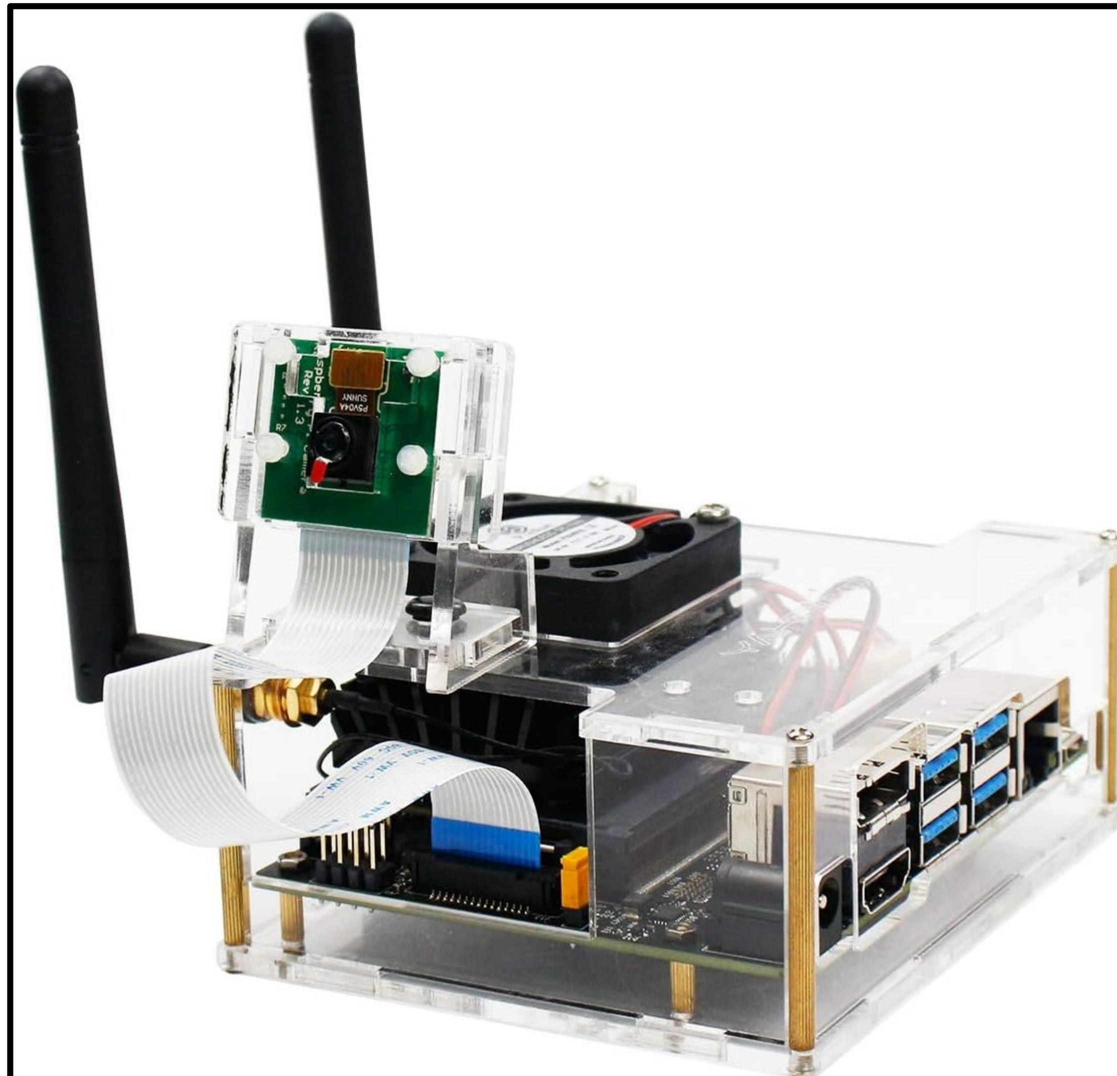


M2 slot for WiFi card



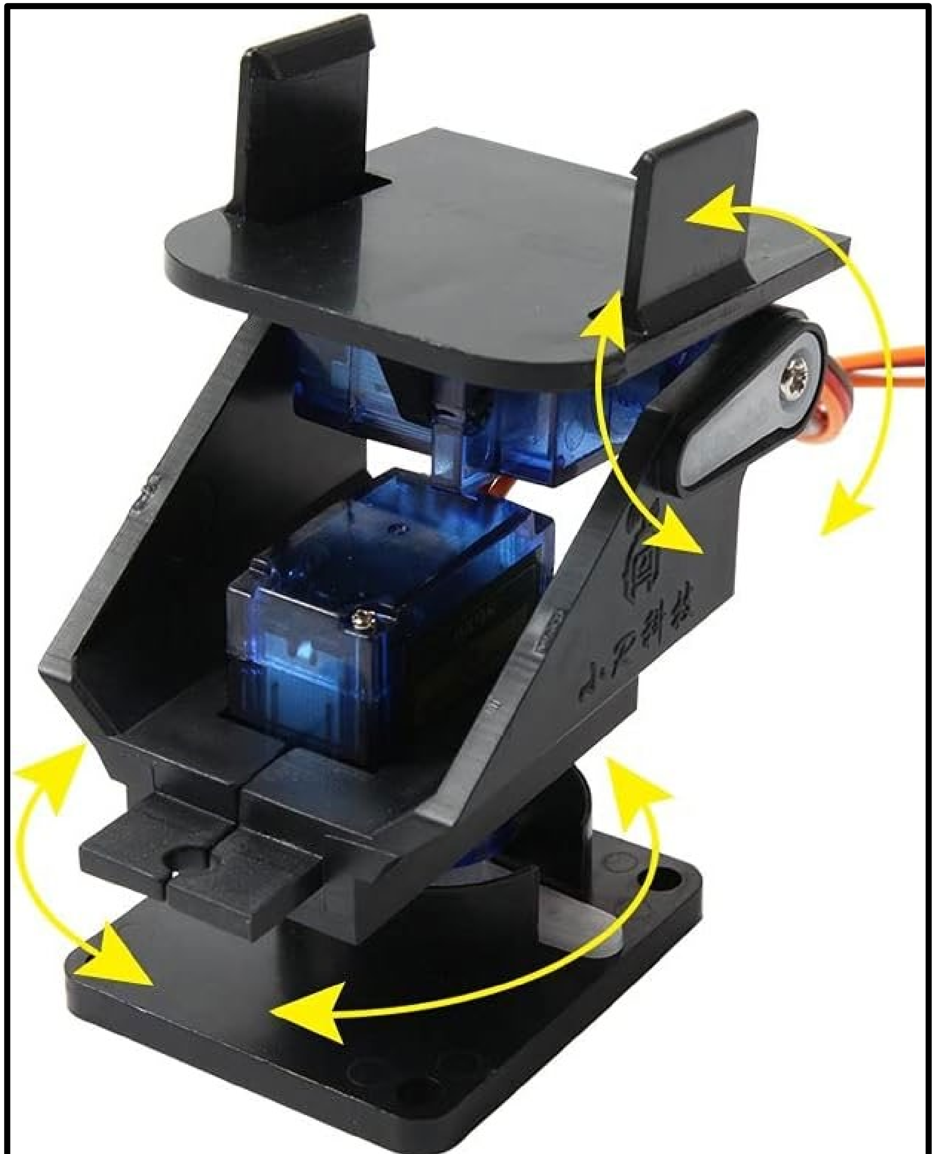
Jetson Nano Acrylic Case

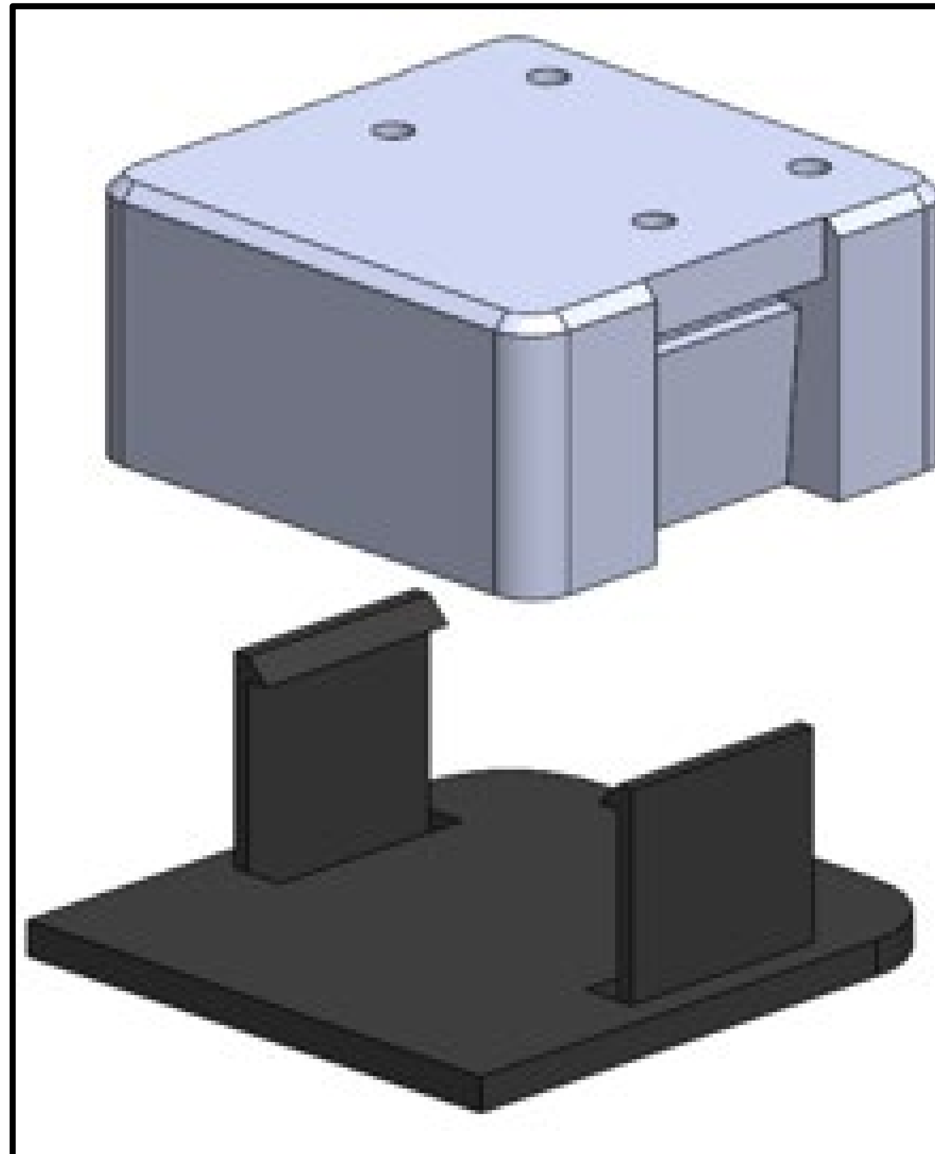




Yahboom Acrylic Case for NVIDIA Jetson Nano

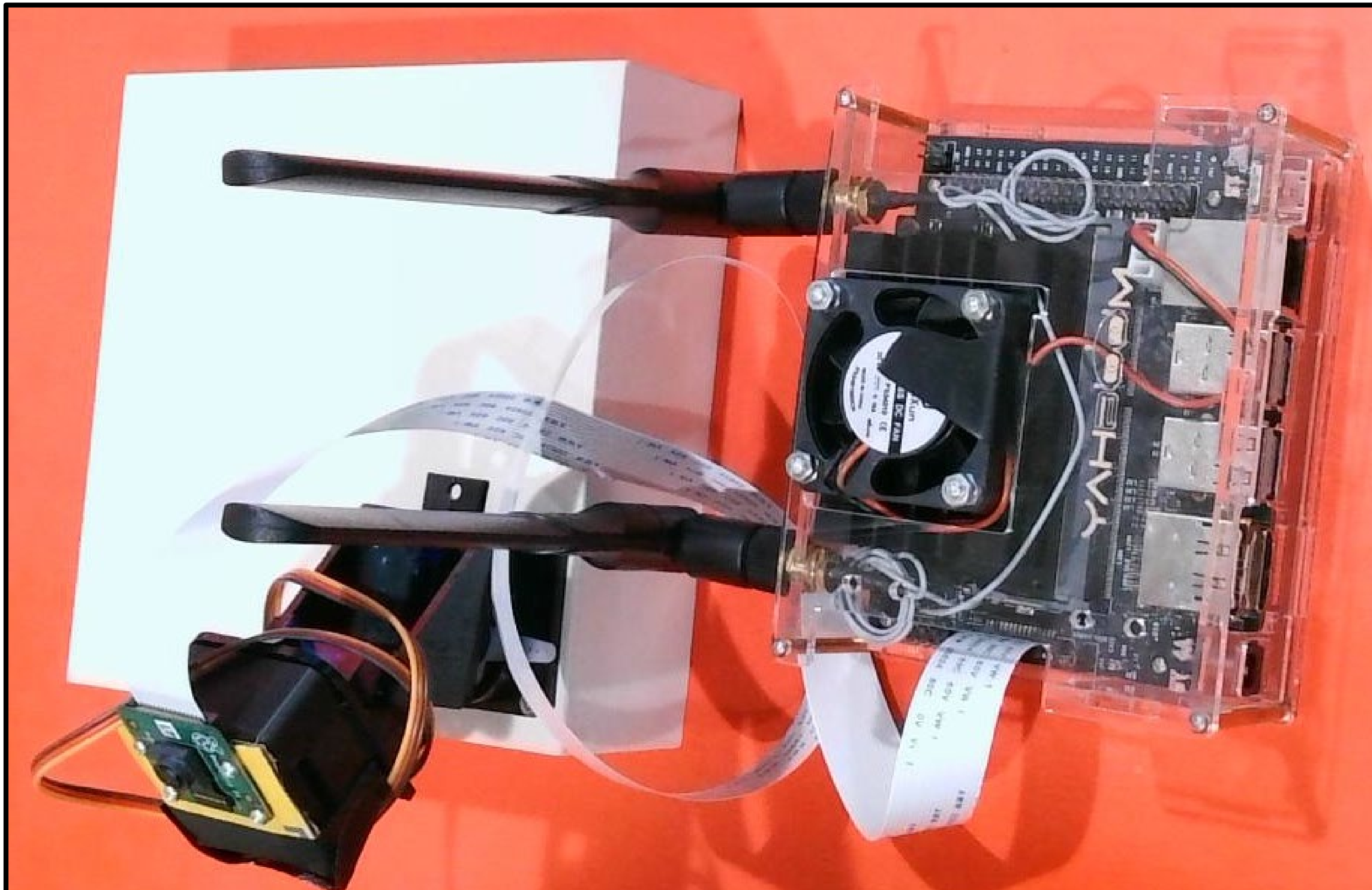
Pan-Tilt Camera Platform





3D-printed adapter for R-PI Holder

Jetson Nano Acrylic Case



Jetson Nano housed in an acrylic case, interfaced to a CSI Camera, mounted on a Pan-Tilt Stage

Jetson Nano: Tech. Specs.

Technical Specifications

GPU	128-core Maxwell
CPU	Quad-core ARM A57 @ 1.43 GHz
Memory	4 GB 64-bit LPDDR4 25.6 GB/s
Storage	microSD (not included)
Video Encode	4K @ 30 | 4x 1080p @ 30 | 9x 720p @ 30 (H.264/H.265)
Video Decode	4K @ 60 | 2x 4K @ 30 | 8x 1080p @ 30 | 18x 720p @ 30 (H.264/H.265)
Camera	1x MIPI CSI-2 DPHY lanes
Connectivity	Gigabit Ethernet, M.2 Key E
Display	HDMI 2.0 and eDP 1.4
USB	4x USB 3.0, USB 2.0 Micro-B
Others	GPIO, I^2C, I^2S, SPI, UART
Mechanical	100 mm x 80 mm x 29 mm

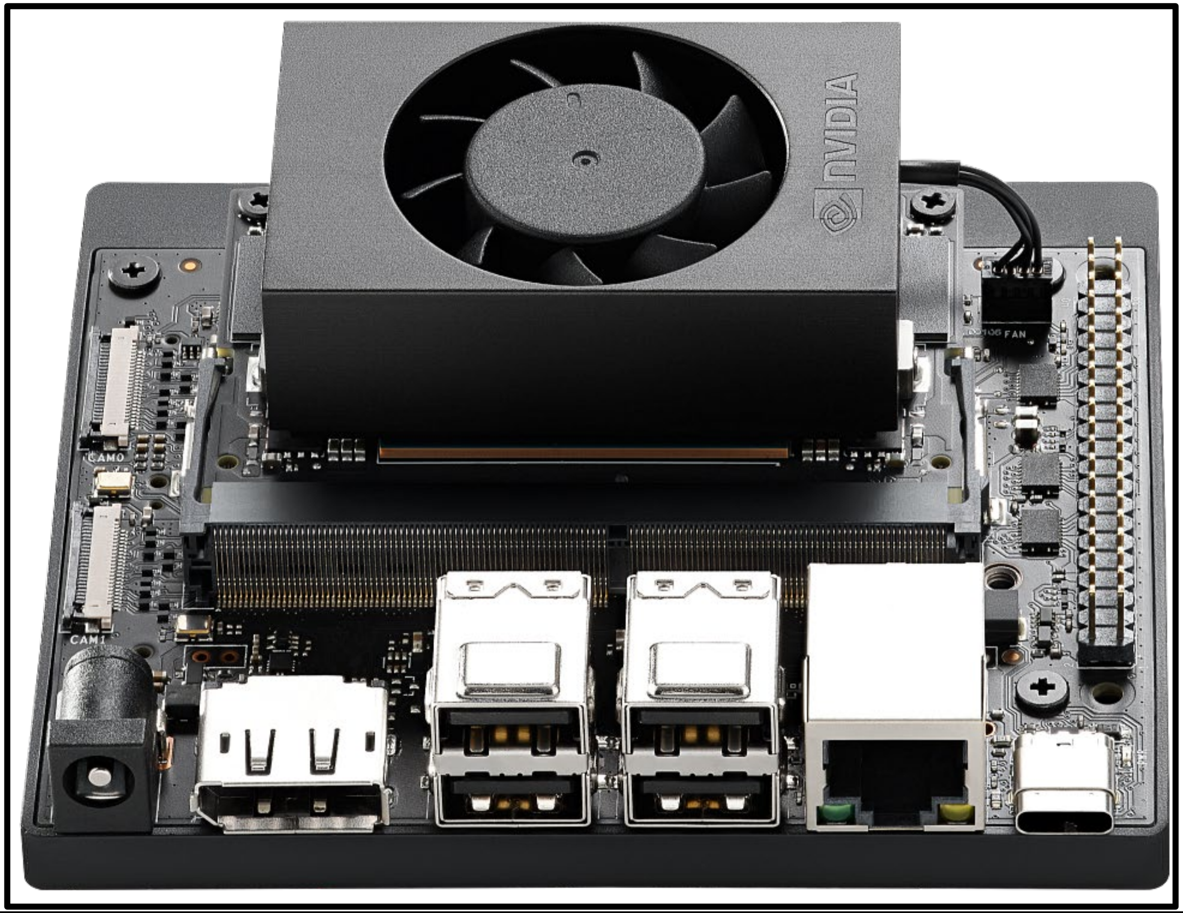
Jetson Nano: Libraries and APIs

- TensorRT and cuDNN for high-performance deep learning applications
- CUDA for GPU accelerated applications across multiple domains
- Multimedia API package for camera applications and sensor driver development
- VisionWorks and OpenCV for visual computing applications

JetPack component	Sample locations on reference filesystem
TensorRT	/usr/src/tensorrt/samples/
cuDNN	/usr/src/cudnn_samples_<version>/
CUDA	/usr/local/cuda-<version>/samples/
Multimedia API	/usr/src/tegra_multimedia_api/
VisionWorks	/usr/share/visionworks/sources/samples/ /usr/share/visionworks-tracking/sources/samples/ /usr/share/visionworks-sfm/sources/samples/
OpenCV	/usr/share/OpenCV/samples/

Jetson Orin Nano Super Developer Kit

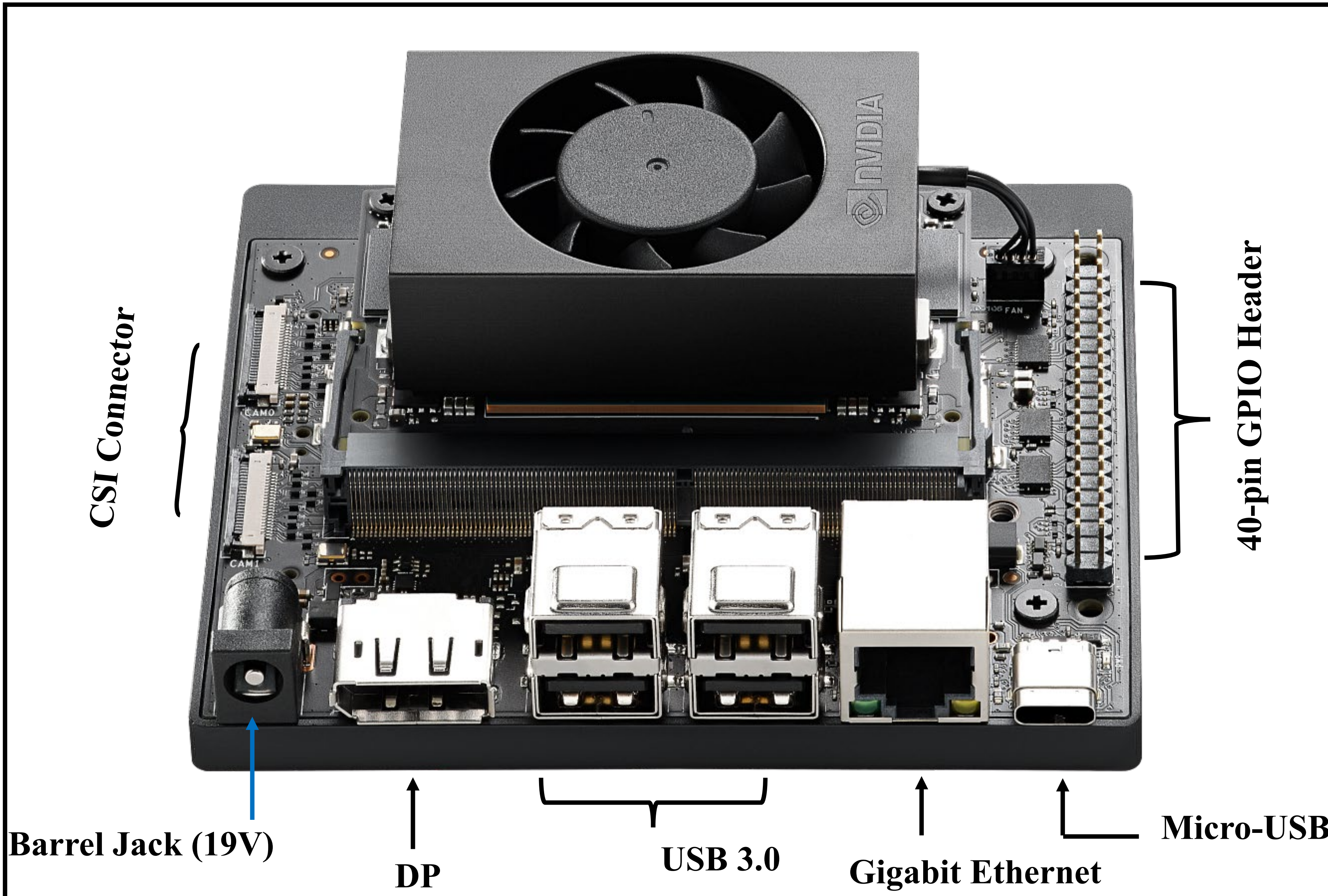
Jetson Orin Nano Super Developer Kit



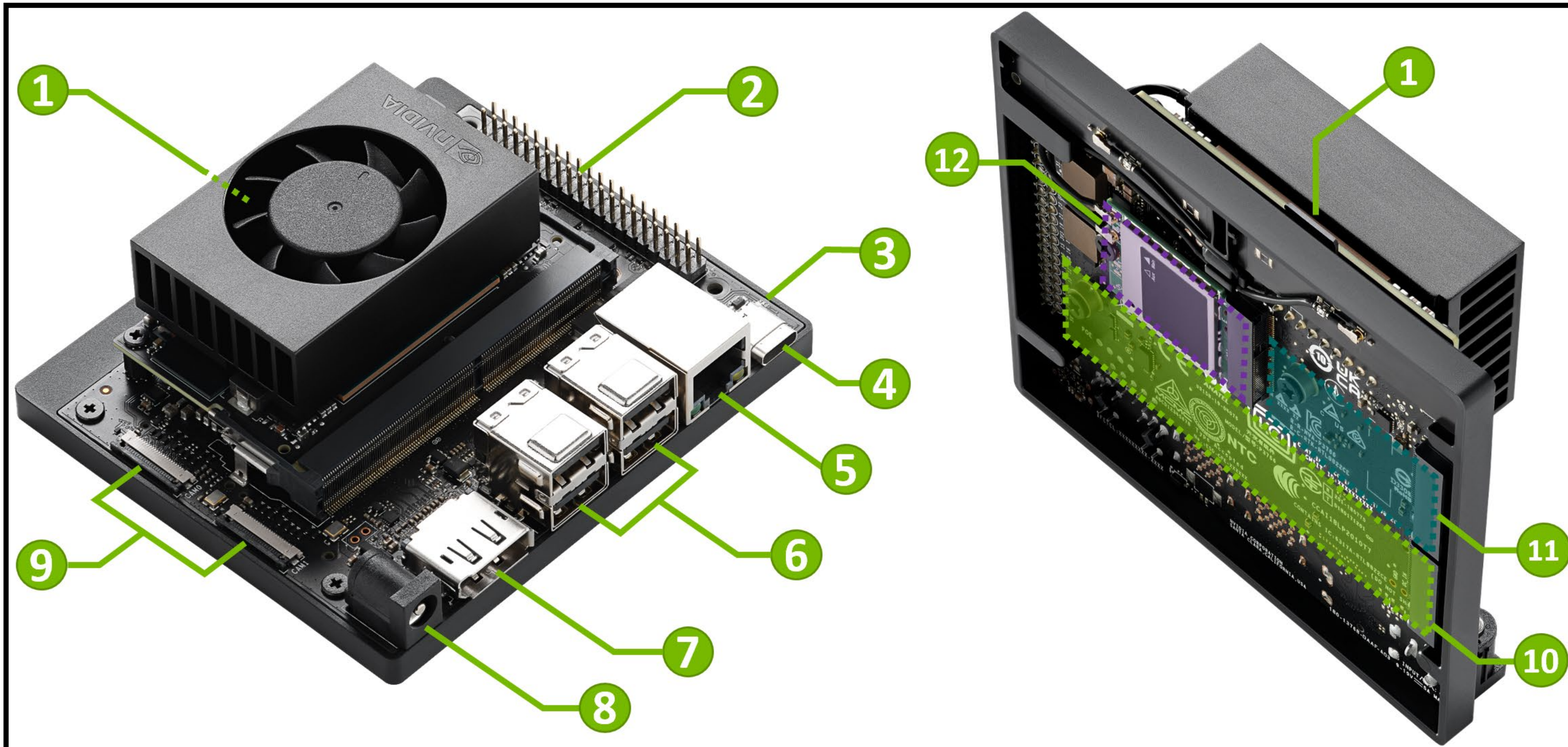
- **CPU: 6-core Arm Cortex-A78AE v8.2, 64-bit, 1.7 GHz.**
- **GPU: Ampere architecture, 1024 CUDA cores, 32 Tensor cores, 1020 MHz**
- **TFLOPS: 67 (INT8, Sparse) TOPS | 33 (INT 8, Dense) TOPS | 17 (FP16) TFLOPS**
- **Memory: 8GB 128-bit**
- **Power: 7W | 15 W | 25W**
- **Size: 103mm × 90mm × 35mm**

Released: March 2023

Jetson Orin Nano Kit Anatomy—I

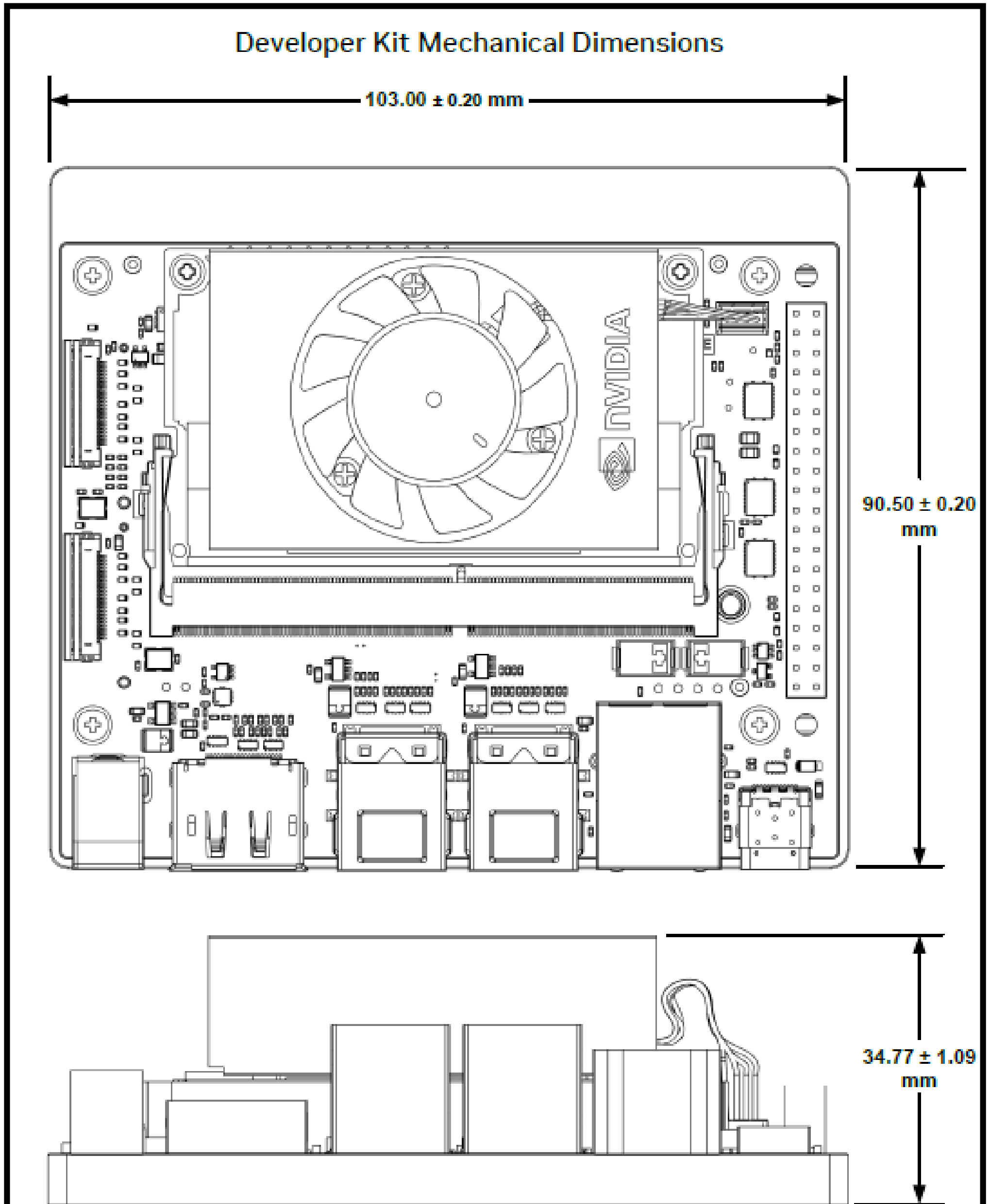


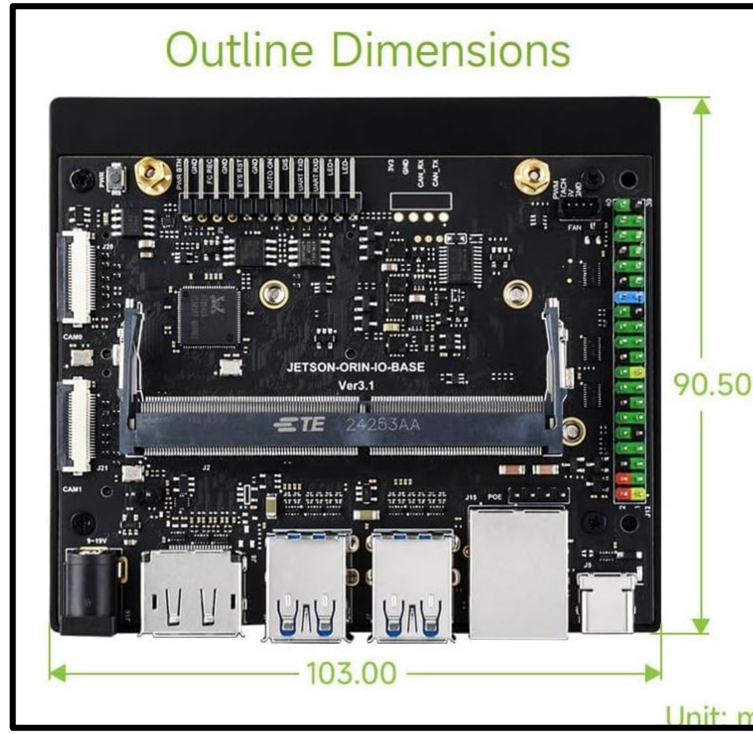
Jetson Orin Nano Kit Anatomy—II



1	microSD card slot	5	Gigabit Ethernet Port	9	MIPI CSI Camera Connectors (x2)
2	40-pin Expansion Header	6	USB 3.2 Type-A ports (x4)	10	M.2 Slot (Key-M, Type 2280)
3	Power Indicator LED	7	DisplayPort Output Connector	11	M.2 Slot (Key-M, Type 2230)
4	USB-C port	8	DC Power Jack	12	M.2 Slot (Key-E, Type 2230)

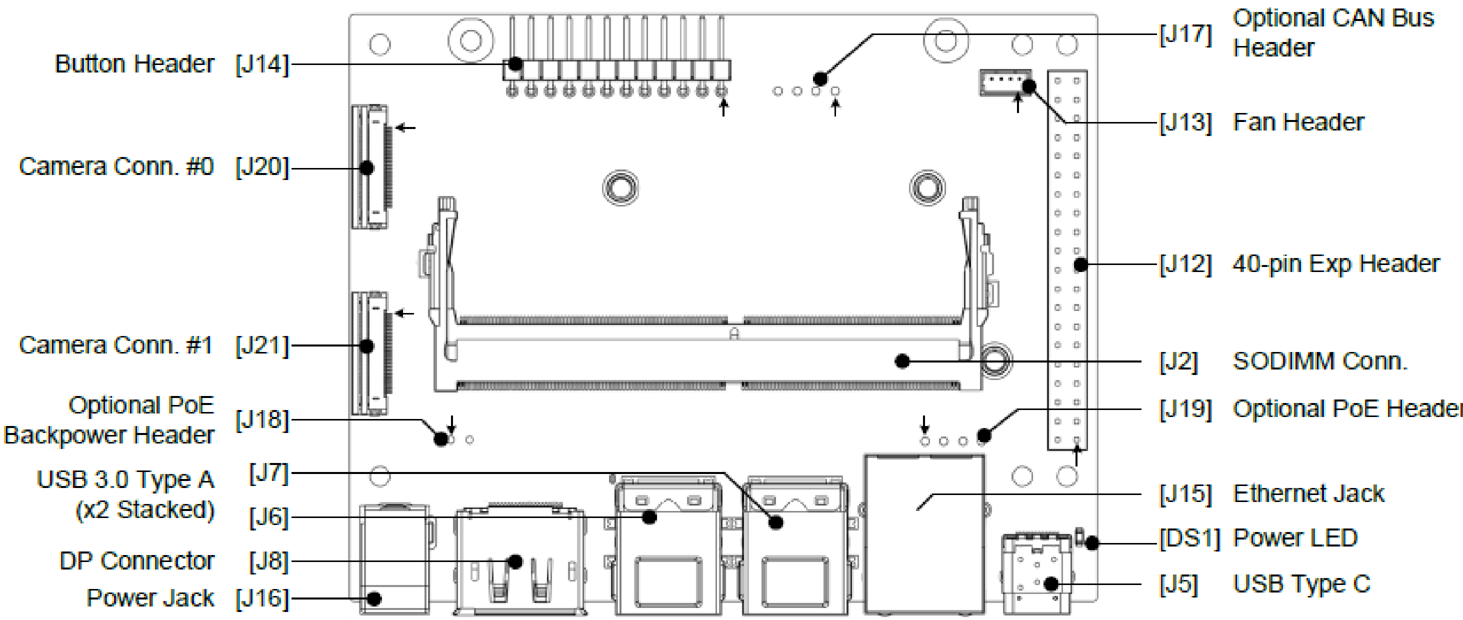
Jetson Orin Nano Kit Dimensions





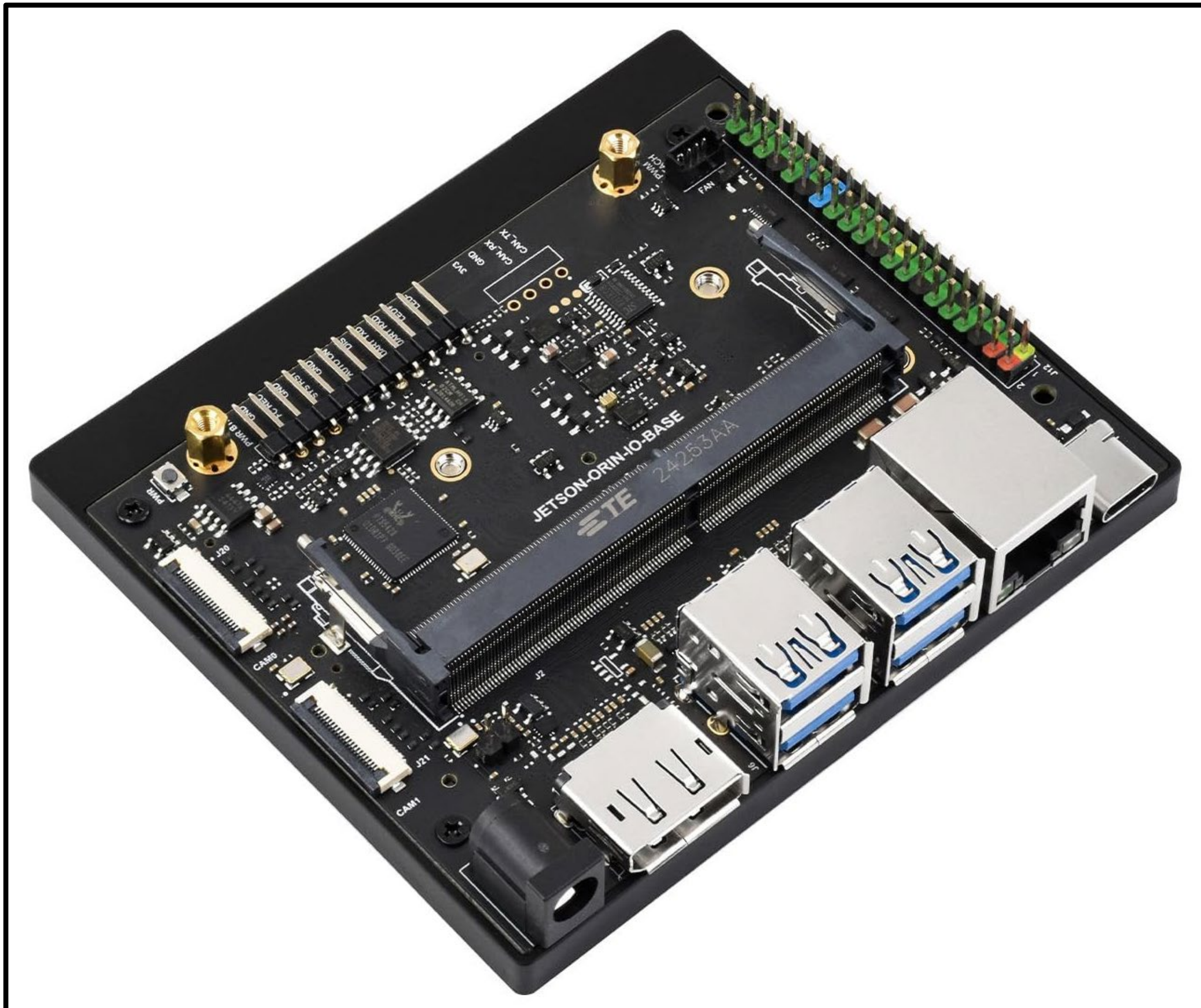
Jetson Orin Nano Carrier Board—I

Top View



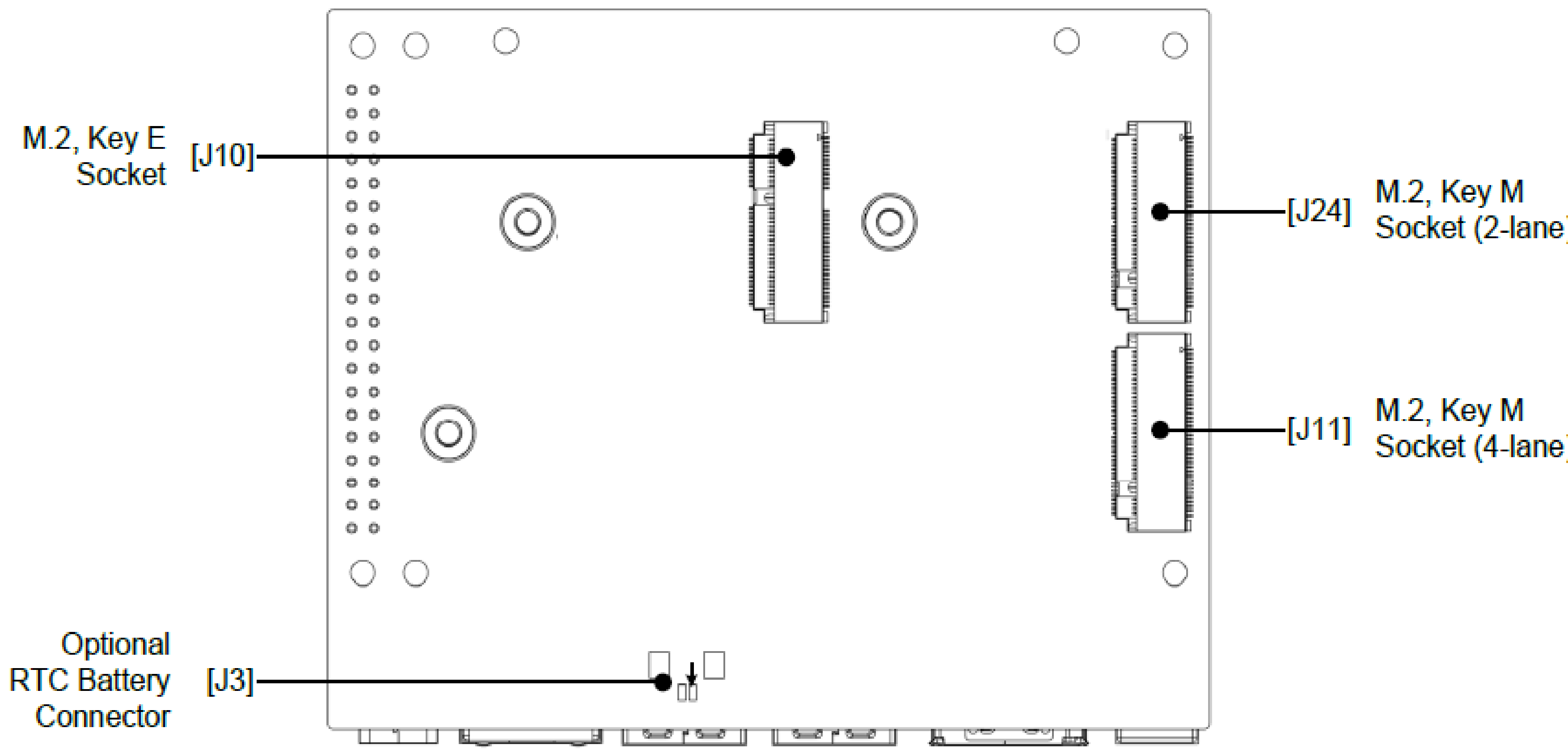
J2	Jetson Orin Nano Conn. (SODIMM, 260-pin)	**J15**	RJ45 Ethernet Socket, 18-pins, RA, Female
J5	USB Type C	**J16**	Power Jack
J6	USB Type A Dual Stacked Connector	**J17**	Optional: CAN Bus Header (1x4, 2.54 mm pitch, RA)
J7	USB Type A Dual Stacked Connector	**J18**	Optional: PoE Backpower Header (1x2, 2.54 mm pitch)
J8	DisplayPort Connector	**J19**	Optional: PoE Header (1x4, 2.54 mm pitch)
J12	40-pin Expansion Header (2x20, 2.54 mm pitch)	**J20**	Camera (#0) Connector (22 pos, 0.5 mm pitch)
J13	Fan Header (4-pin, 1.25 mm pitch)	**J21**	Camera (#1) Connector (22 pos, 0.5 mm pitch)
J14	Button Header (1x12, 2.54 mm pitch, RA)	**DS1**	Power LED (Green)

Jetson Orin Nano Carrier Board—II



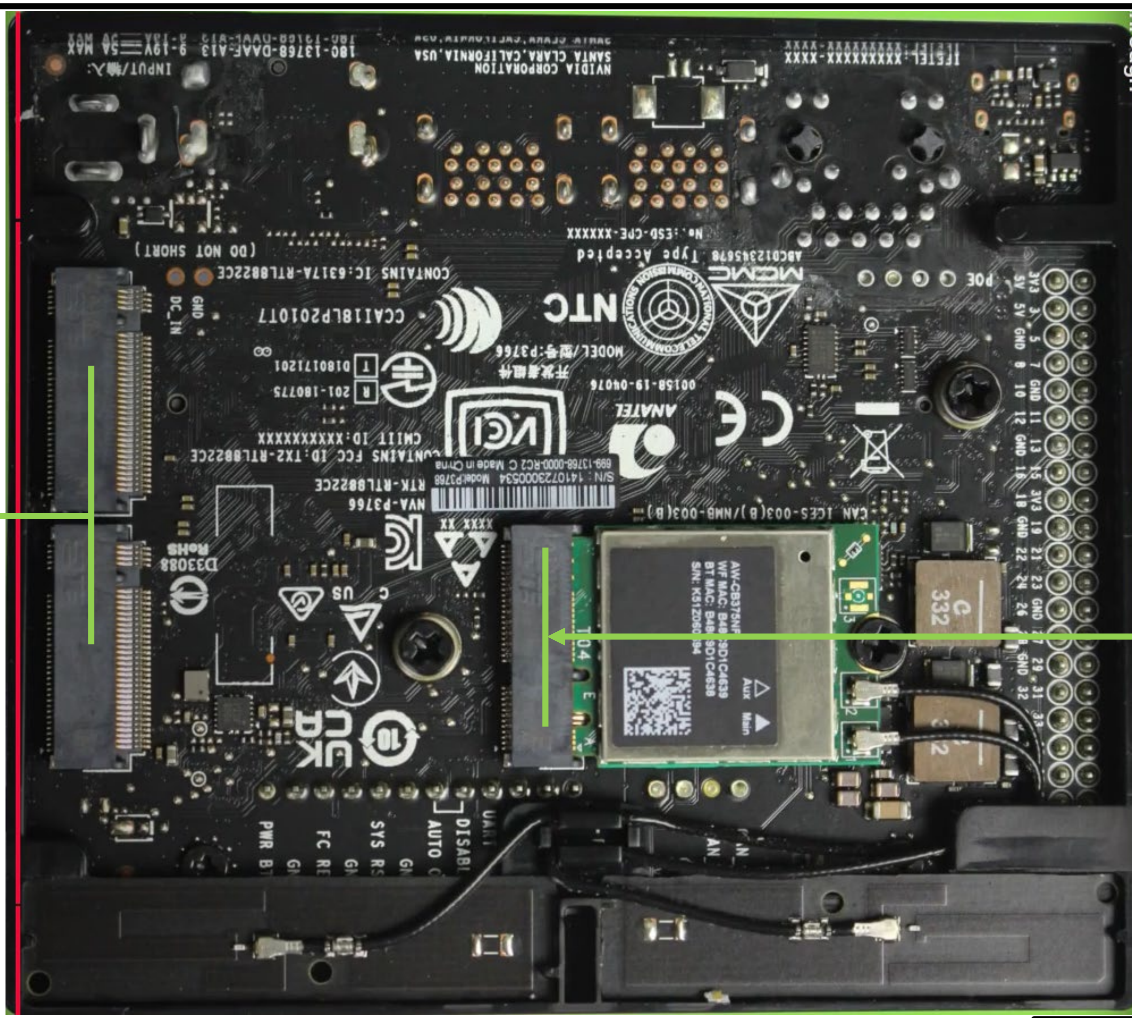
Jetson Orin Nano Carrier Board—III

Bottom View



J3	Optional: RTC Back-up Coin Cell Socket (CR1225)	**J11**	M.2 Key M Slot (75-pin)
J10	M.2 Key E Connectivity Slot (75-pin)	**J24**	M.2 Key M Slot (75-pin)

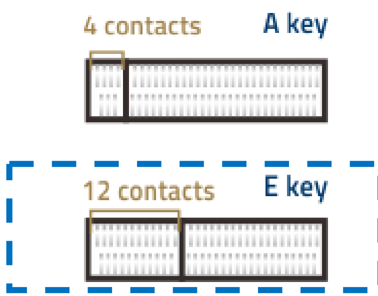
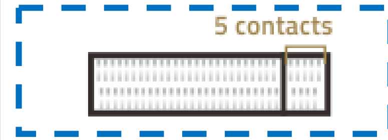
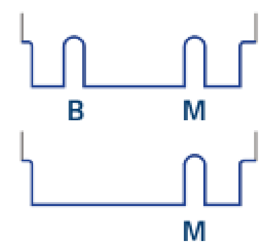
Jetson Orin Nano Carrier Board—IV



Flexible Storage And Wireless Expansion

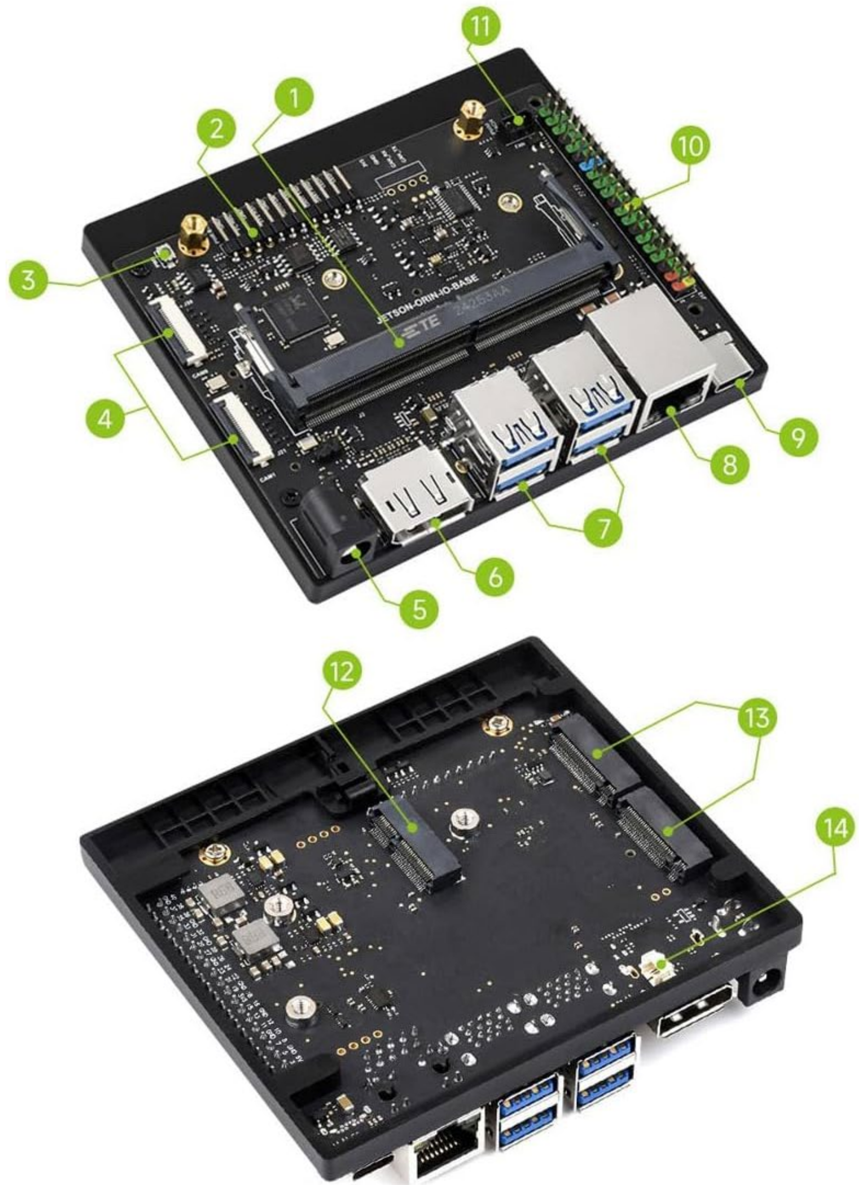
Onboard 2× M.2 Key M Ports For Easy Connecting Solid State Drives And 1× M.2 Key E Interface For Connecting Wireless NIC, Which Can Reduce Cable Connection

Aside: M.2 Key Interfaces

	A or E key	B key	M key
Motherbord's M.2 Slot Type			
Supported Interface	PCIe(x1)/ USB2.0	SATA/ PCIe(x2)/ USB2.0/ USB3.0	PCIe(x4)/ SATA**
Supported M.2 SSD Pin Patterns	A+E key	B+M key	B+M key / M key
			
Max SSD Transfer Rate*	1 GB/s	2 GB/s	8 GB/s
Common Applications	Wireless Connectivity incl. Wi-Fi, Bluetooth	SSD/ WWAN+GNSS	SSD/ Host Interface

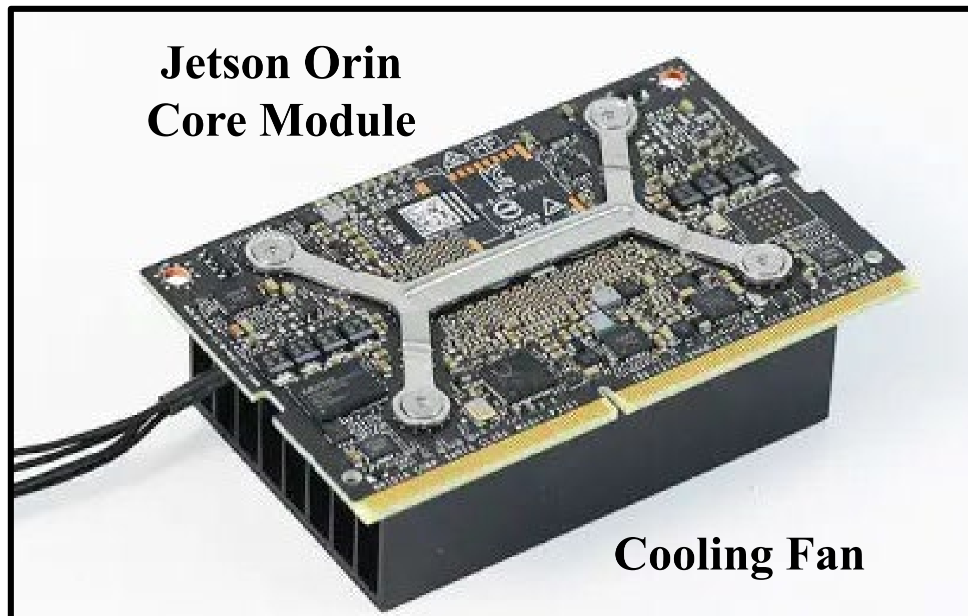
Jetson Orin Nano Carrier Board—V

Jetson Orin Nano Carrier Board—VI

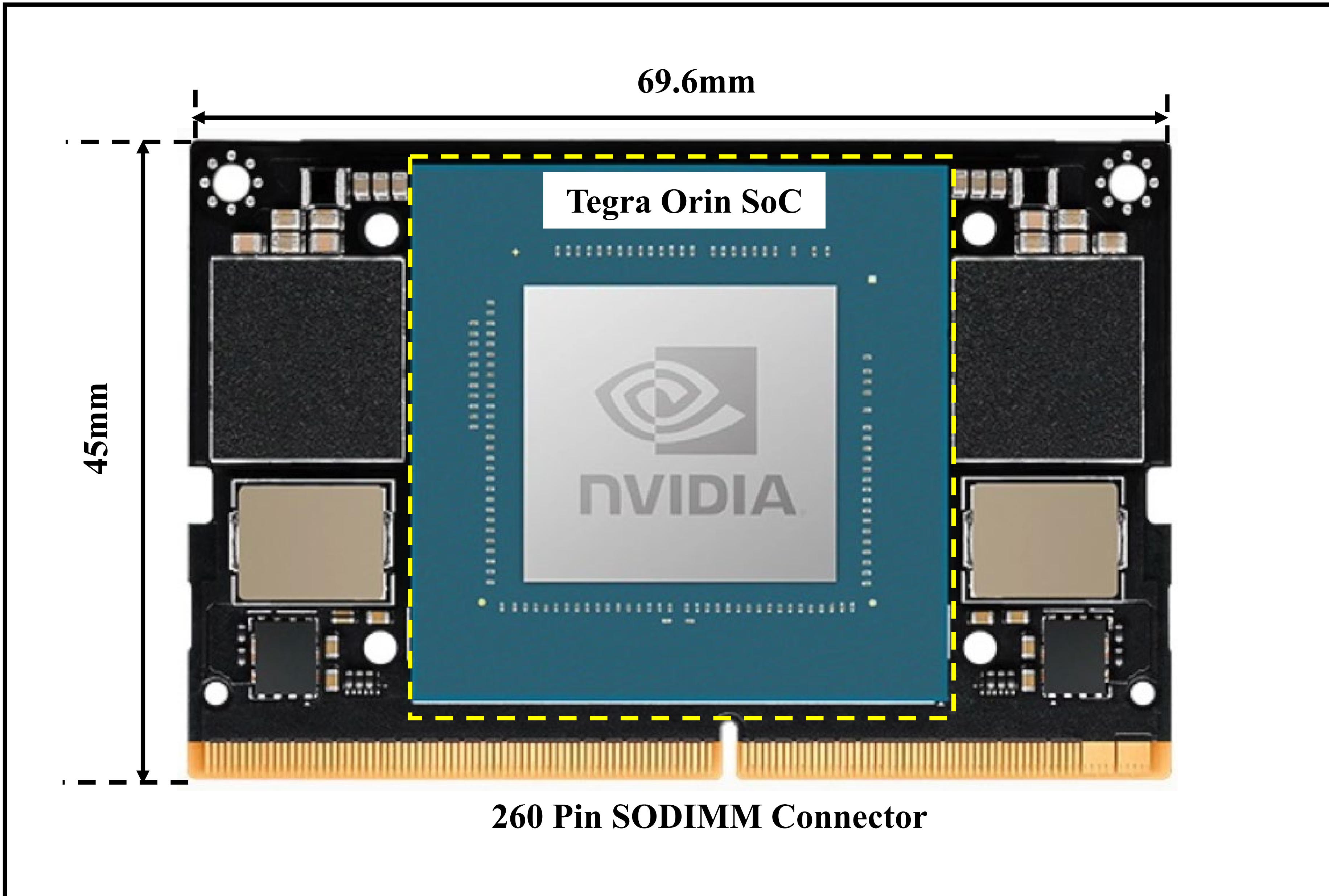


1. **260PIN DO-DIMM socket**
   for connecting Jetson Orin Nano/NX module
2. **12PIN header**
   Adapting pins for power supply, reset button, force recovery setting, serial port debug, auto power-up enable and power indicator functions
3. **Power button**
4. **2× 22PIN CSI camera connectors**
   2× MIPI CSI 4-lane camera connectors
5. **DC Power jack**
   Supports wide range 9V~19V power supply, compatible with 5.5 × 2.5mm male plug
6. **DisplayPort connector**
   Supports 4K display
7. **4× USB 3.2 Gen2 Type A ports**
   Supports up to 10Gbps data transmission
8. **RJ45 Gigabit Ethernet port**
9. **USB Type-C Interface**
   for system burning and data communication
10. **40PIN header**
11. **4PIN PWM fan header**
12. **M.2 Key E**
    for connecting wireless NIC
13. **2x M.2 Key M**
    for connecting NVME solid state drives
14. **RTC backup battery header**
    PH1.25 2P

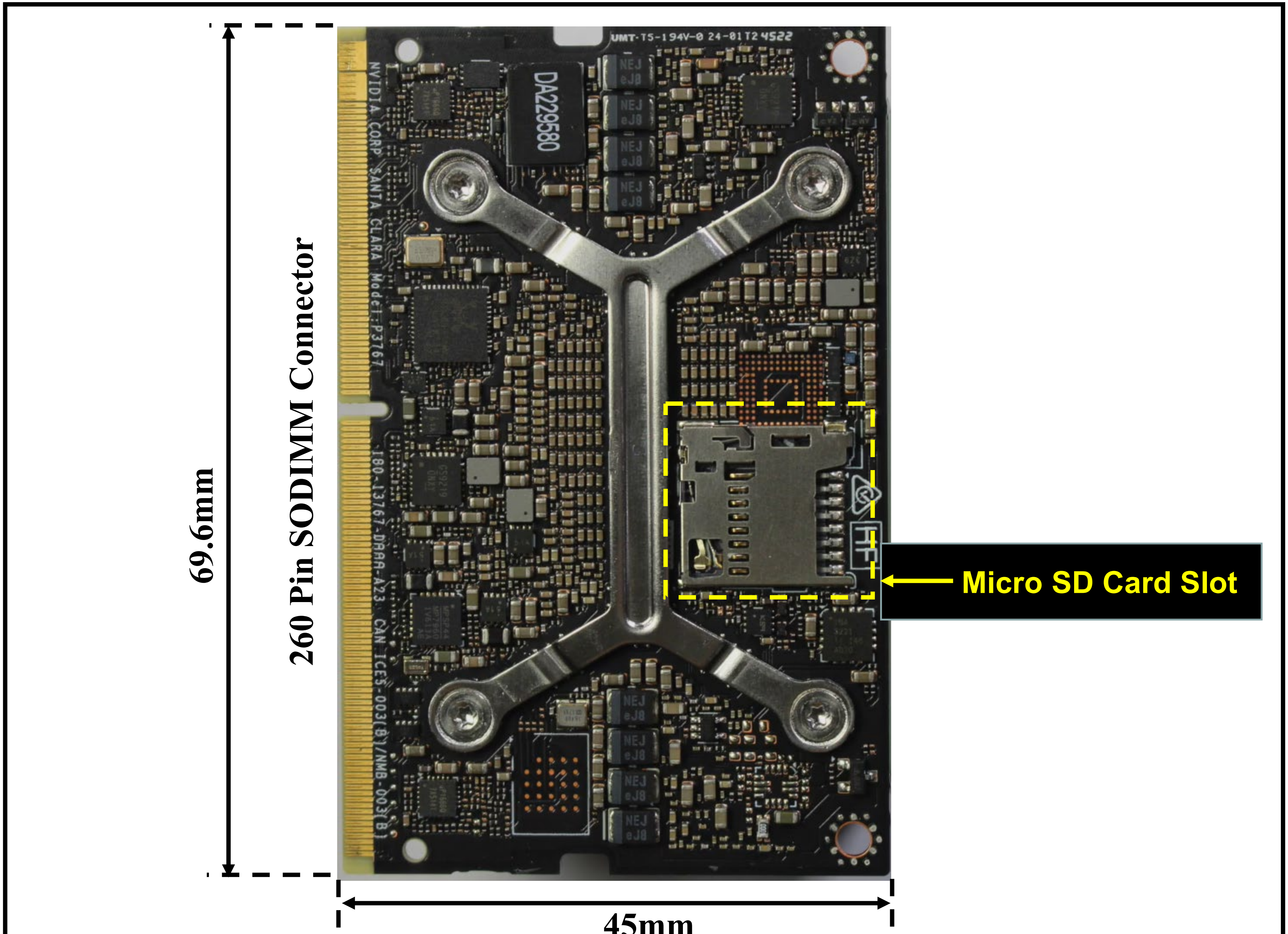
Jetson Orin Nano Core Module + Cooling Fan



Jetson Orin Nano Core Module

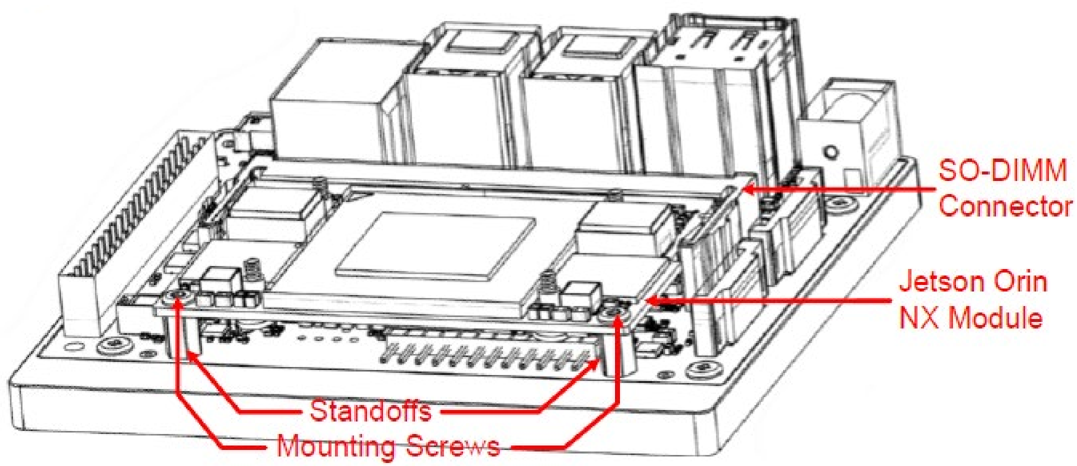


Jetson Orin Nano Core Module

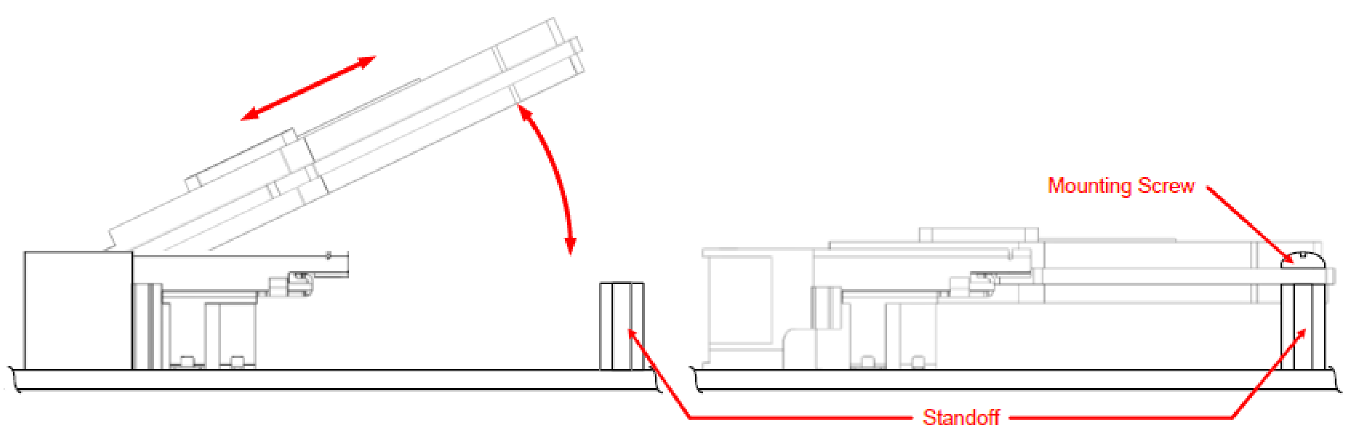


Mount Jetson Orin Module on Carrier Board—I

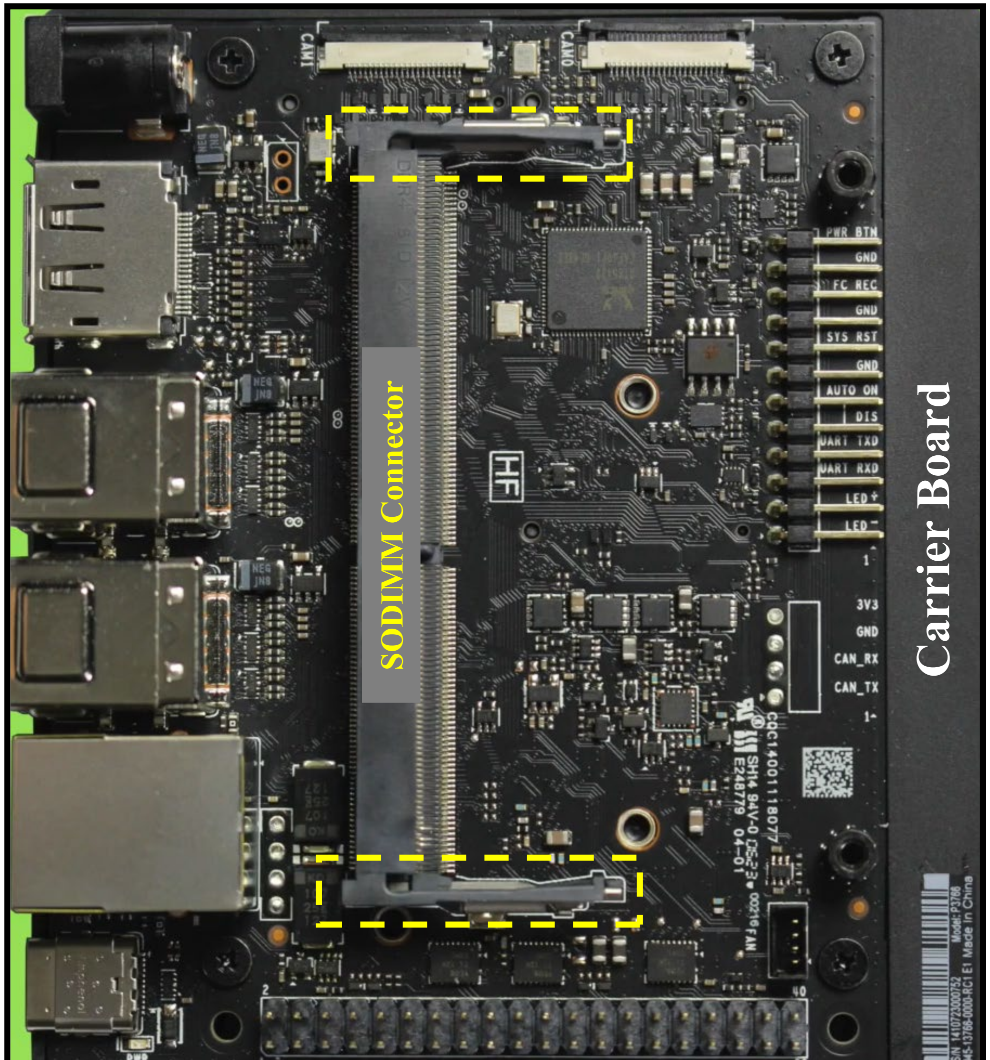
Jetson Orin Module Installed in SODIMM Connector



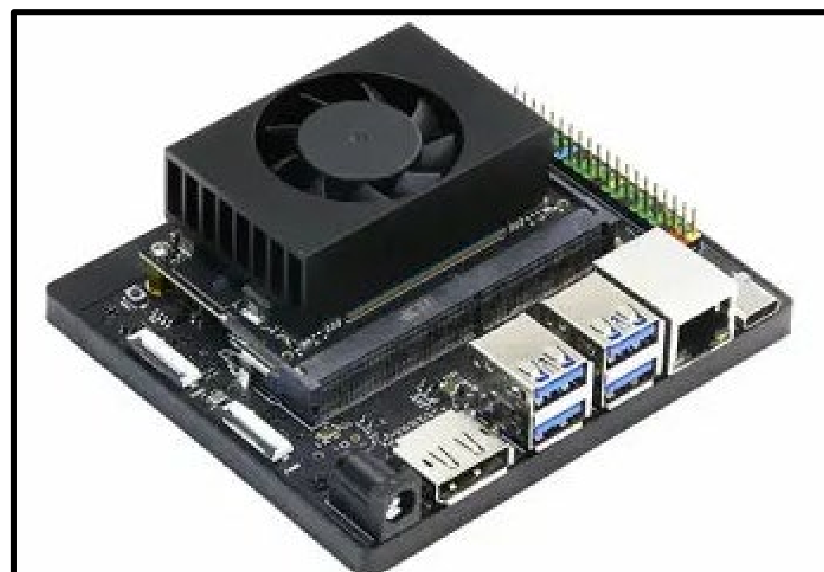
Module to Connector Assembly Diagram



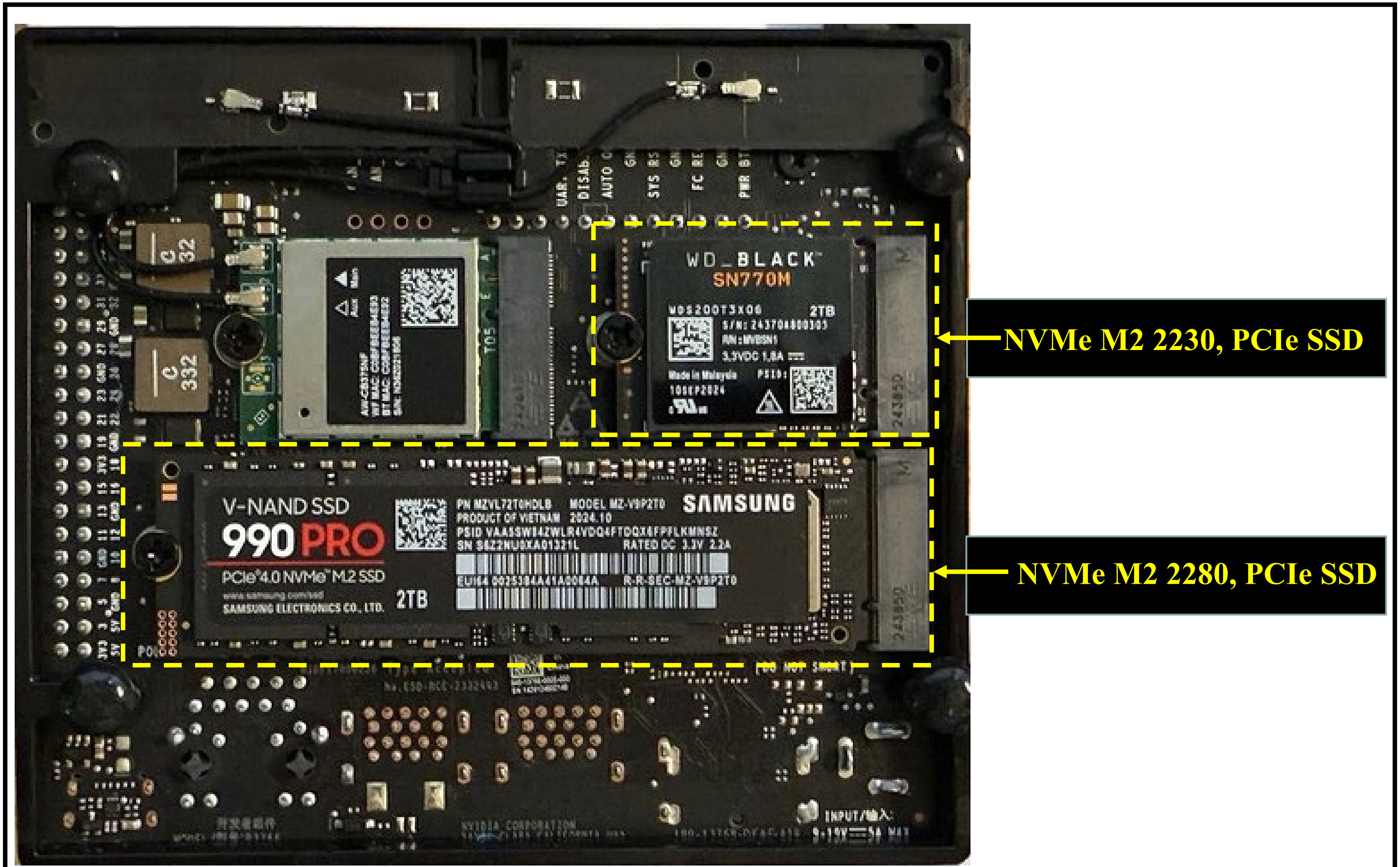
Mount Jetson Orin Module on Carrier Board—II



Core module+fan mounted on Carrier Board using SODIMM Connector

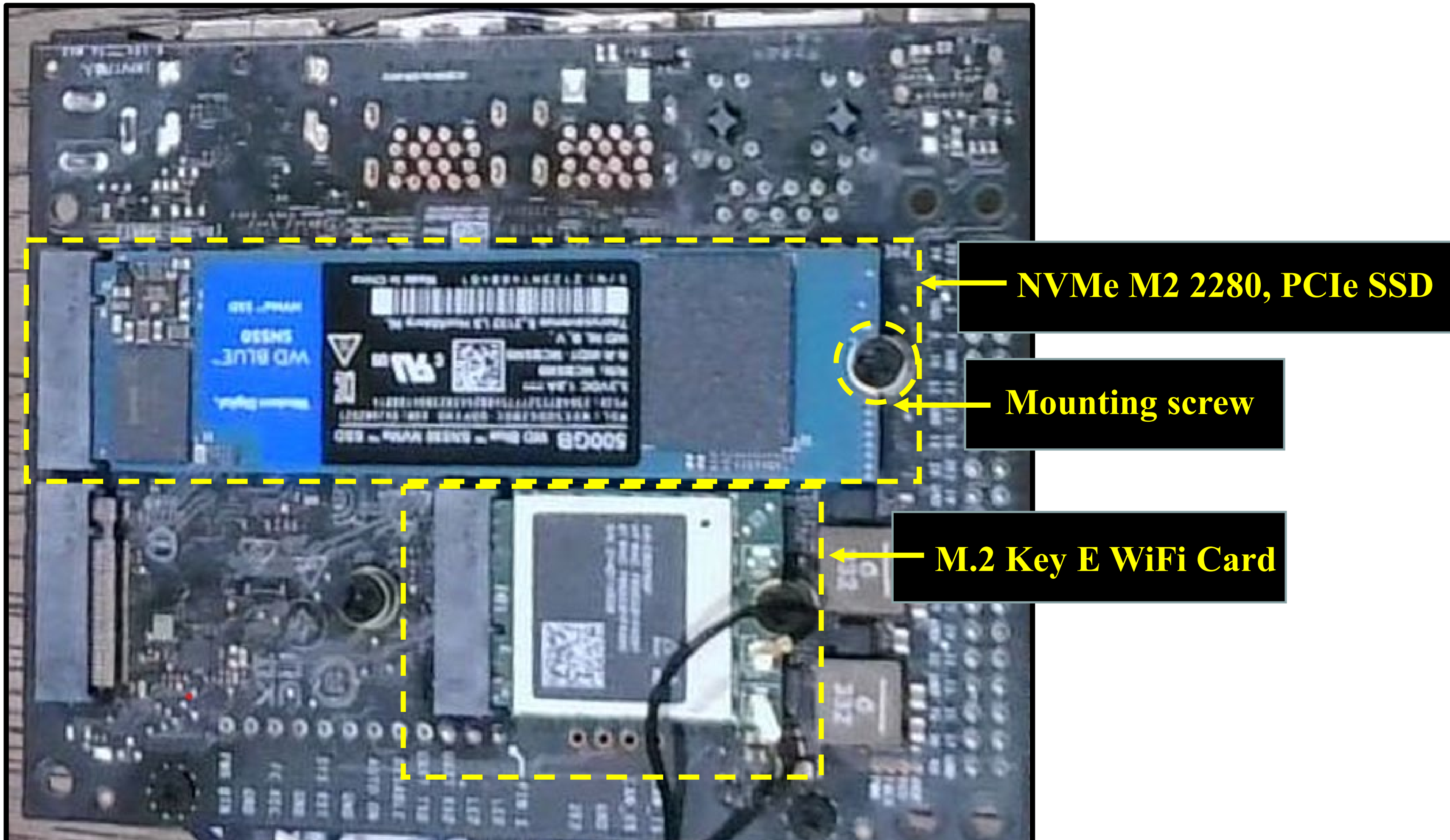


Jetson Orin Nano with NVMe SSD—I



Jetson Orin Nano with NVMe SSD—II

WD PCI SD Card Installed



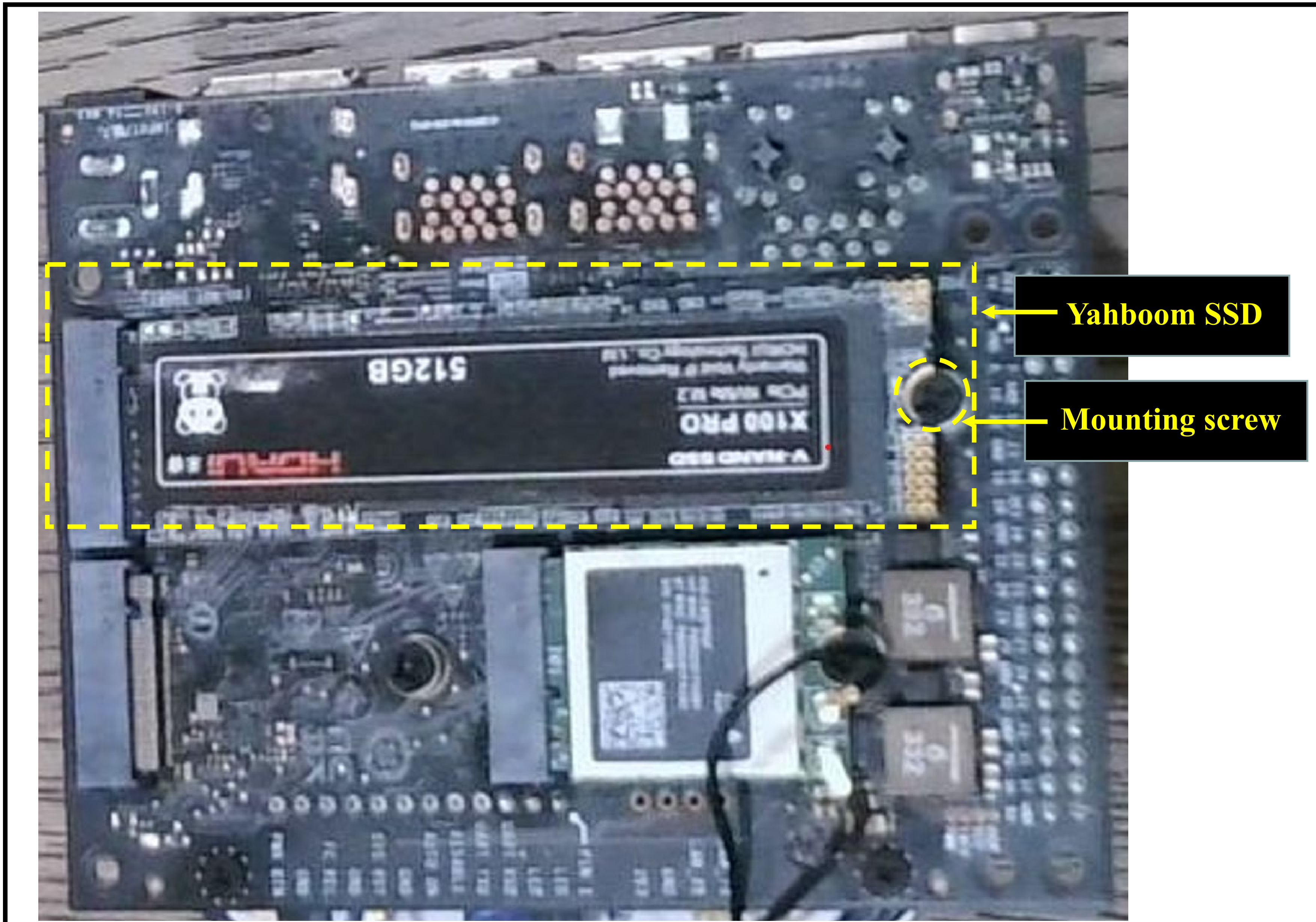
Jetson Orin Nano with NVMe SSD—III

Removed screw, SSD popped up at an angle, gently too it out, see contacts facing up

Insert <u>Yahboom SSD</u> at an angle, slide in the contacts gently to fit in, then push down, and finally mount screw

Jetson Orin Nano with NVMe SSD—IV

Yahboom PCI SSD Installed

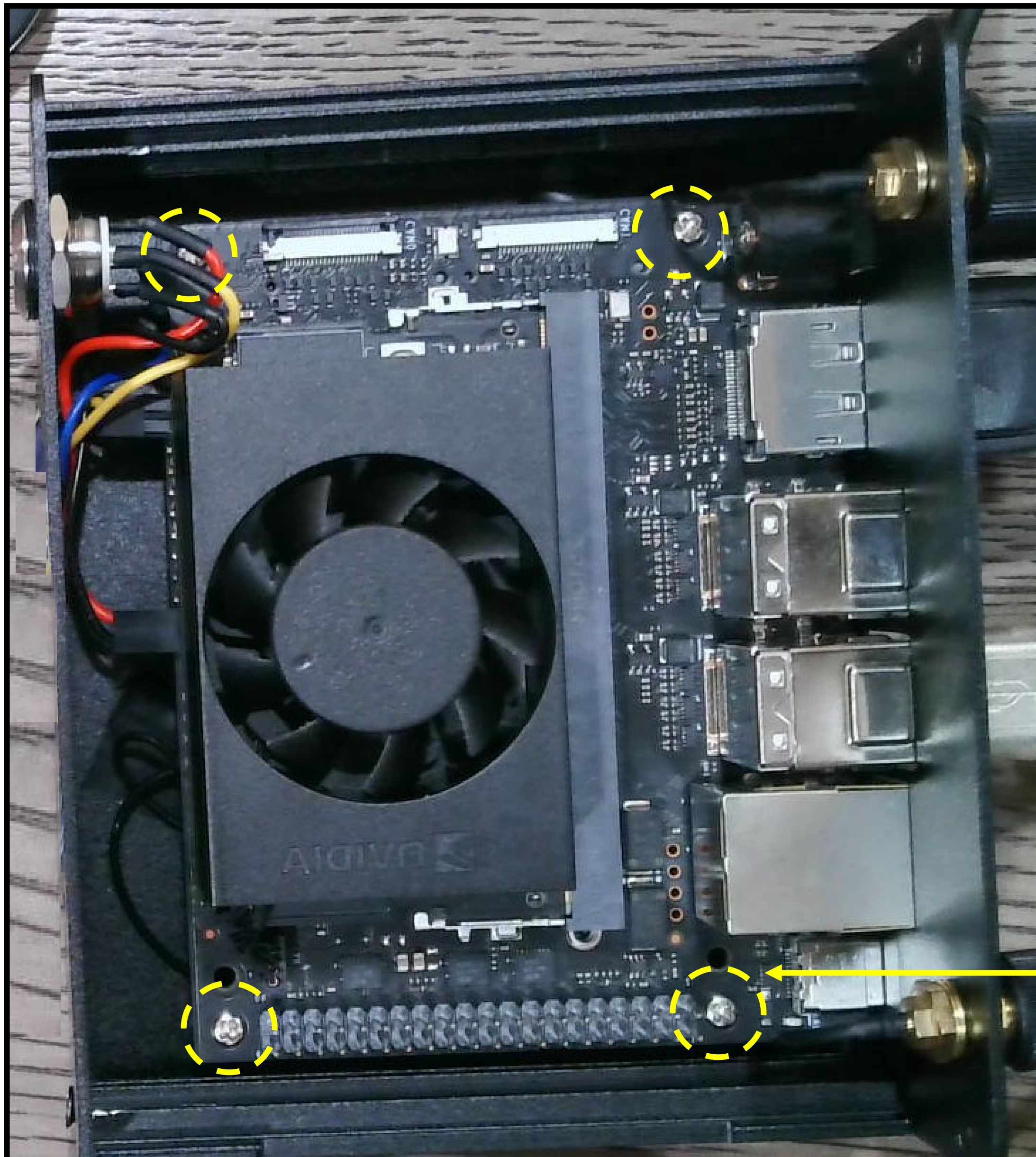


Jetson Orin Nano Waveshare Case—I

Waveshare Case Package Contents

1 Aluminum body case (top and bottom + side plates) × 1
2 Camera holder bottom mount × 1
3 Camera holder top mount × 1
4 Antenna (2PCS) × 1
5 IPEX4 to SMA cable (2PCS) × 1
6 Button with 4PIN wires × 1
7 Flat ribbon cable 2*20PIN ~55mm × 1
8 Screwdriver × 1
9 Screws pack and accessories × 1

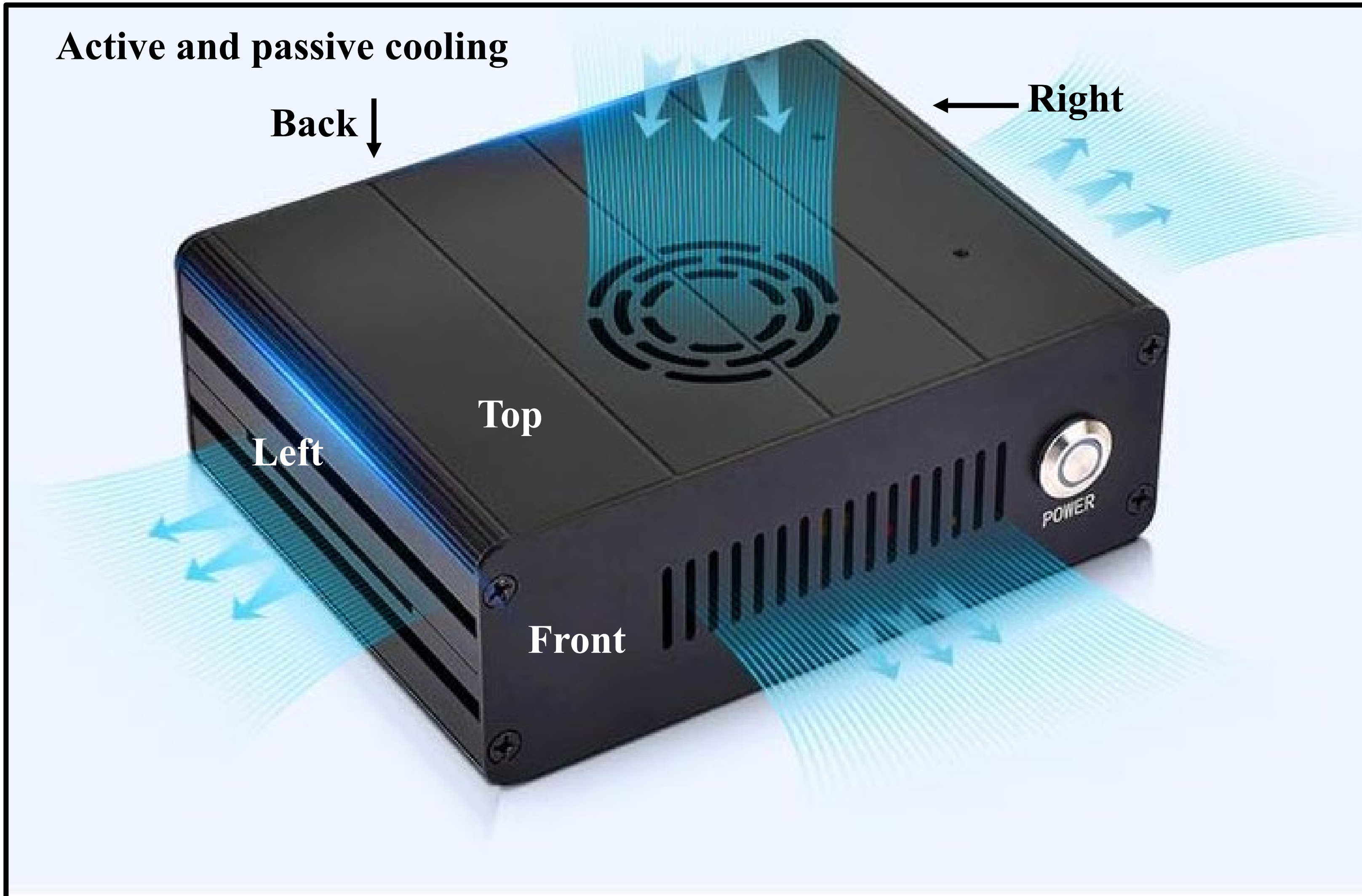
Jetson Orin Nano Waveshare Case—II



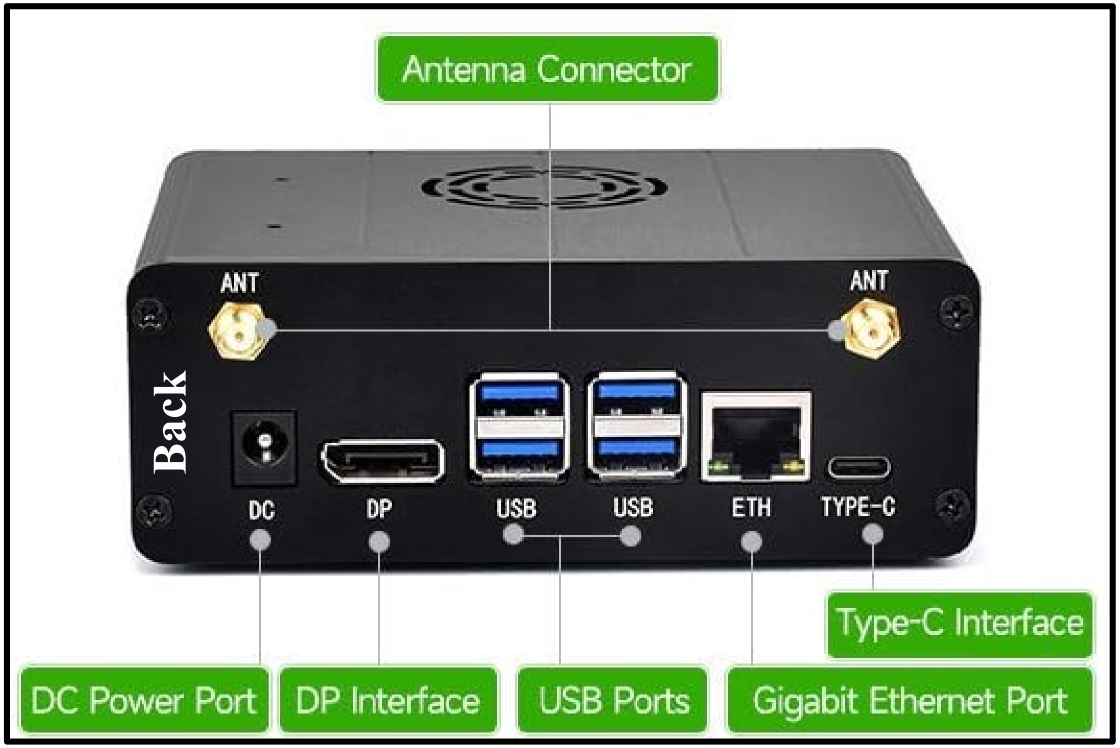
Mounting screws

Jetson Orin Nano Waveshare Case--III

Jetson Orin Nano Waveshare Case



Jetson Orin Nano Waveshare Case

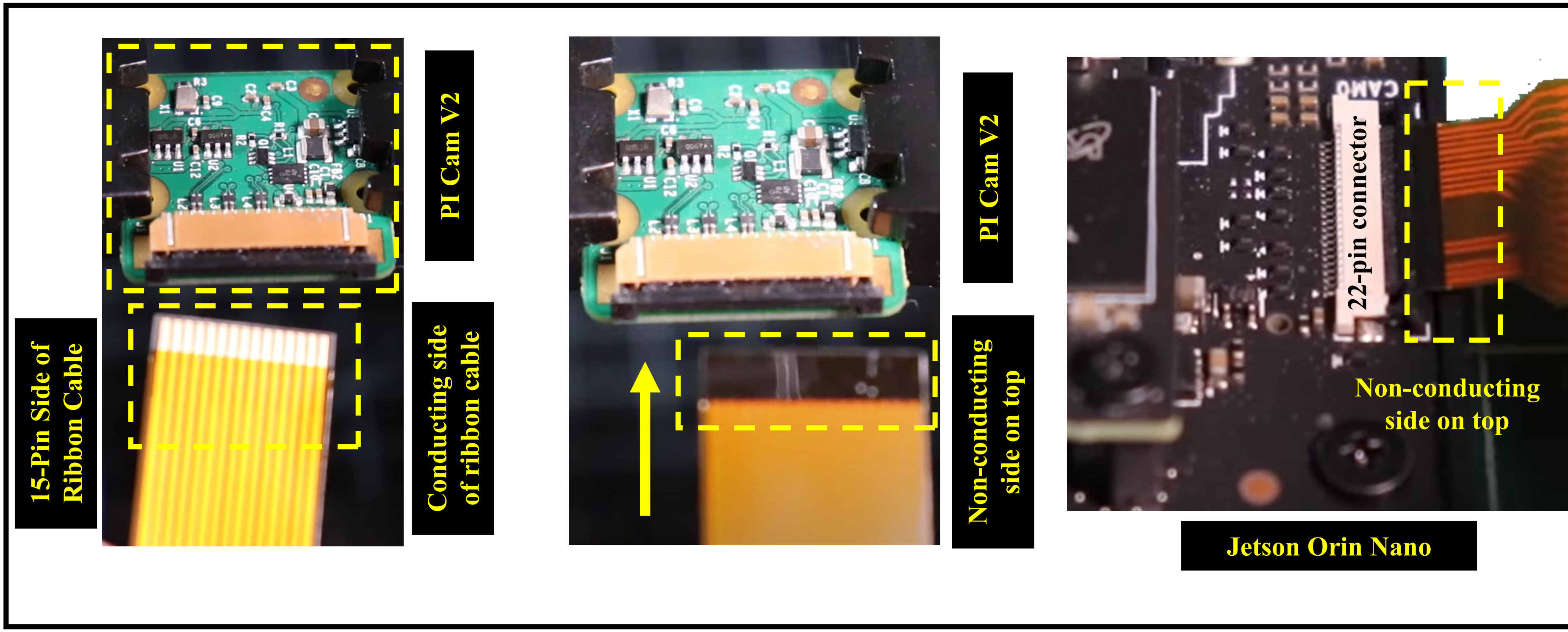


Functional Openings

Waveshare Case Assembly Video

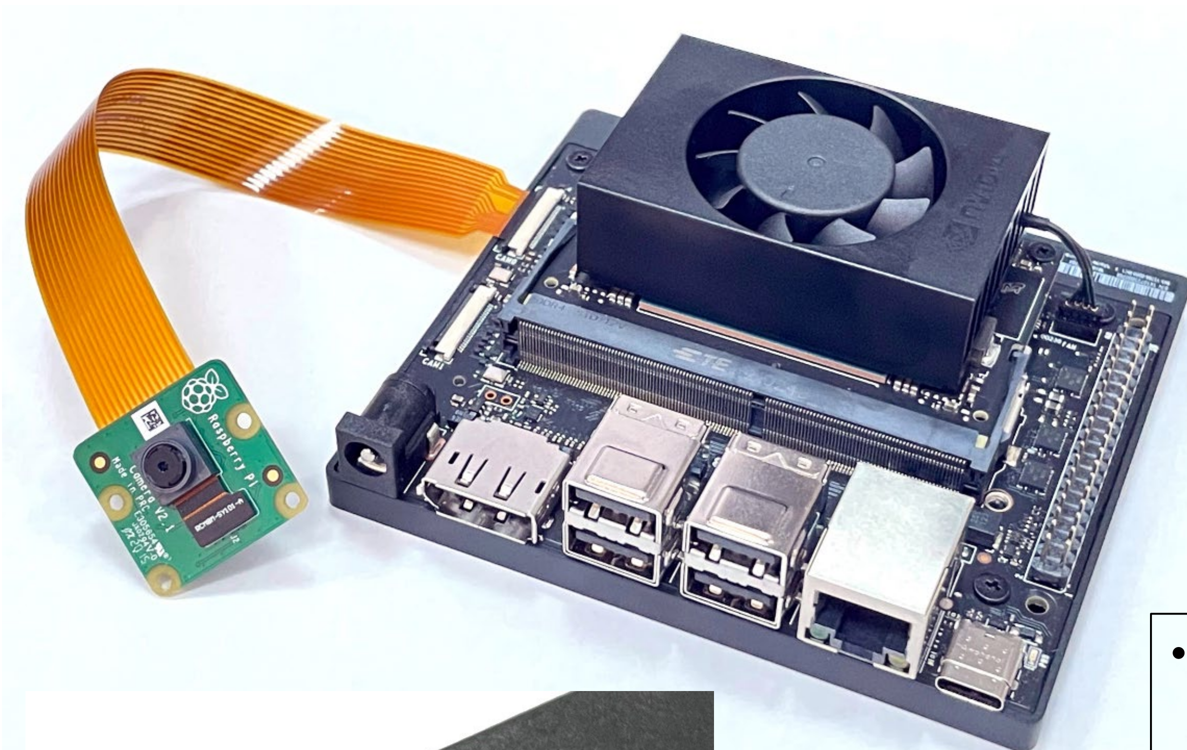
[Assembly Video]

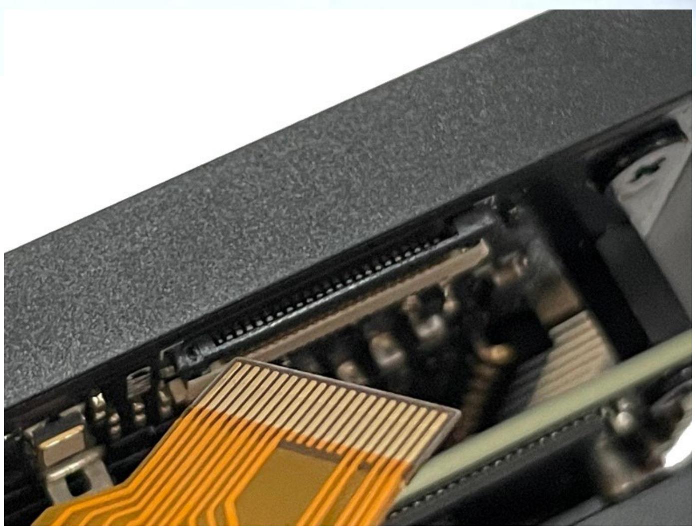
Jetson Orin Nano: Installing CSI Camera



- Camera connector on Jetson Orin Nano: 22 pins
- PI Cam—2: 15 pin connector
- Need 15 to 22 pin cable

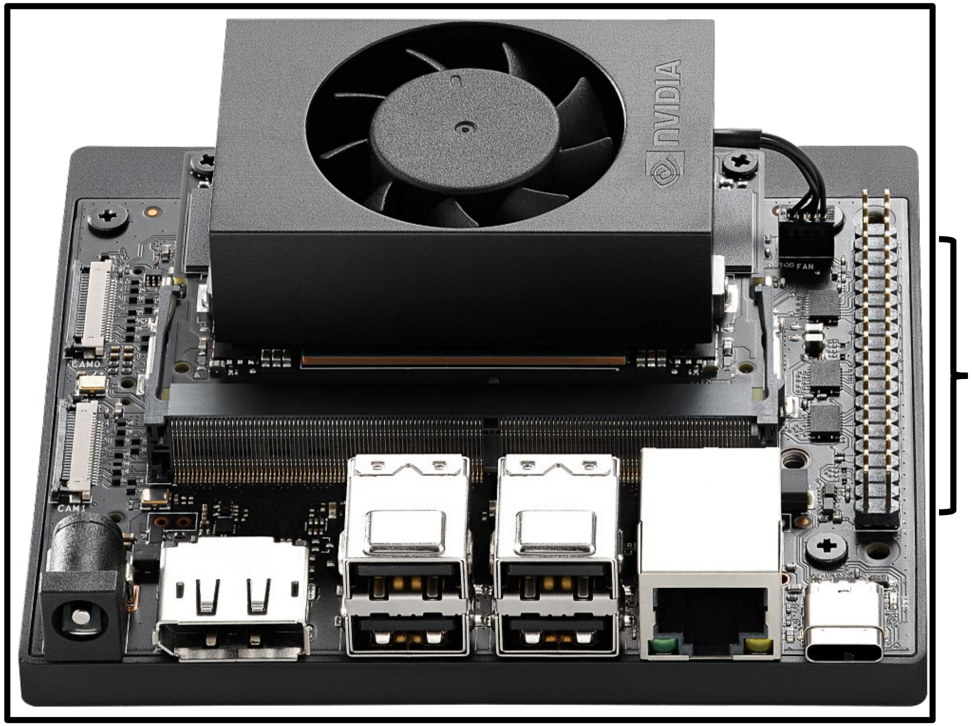
Jetson Orin Nano: Installing CSI Camera





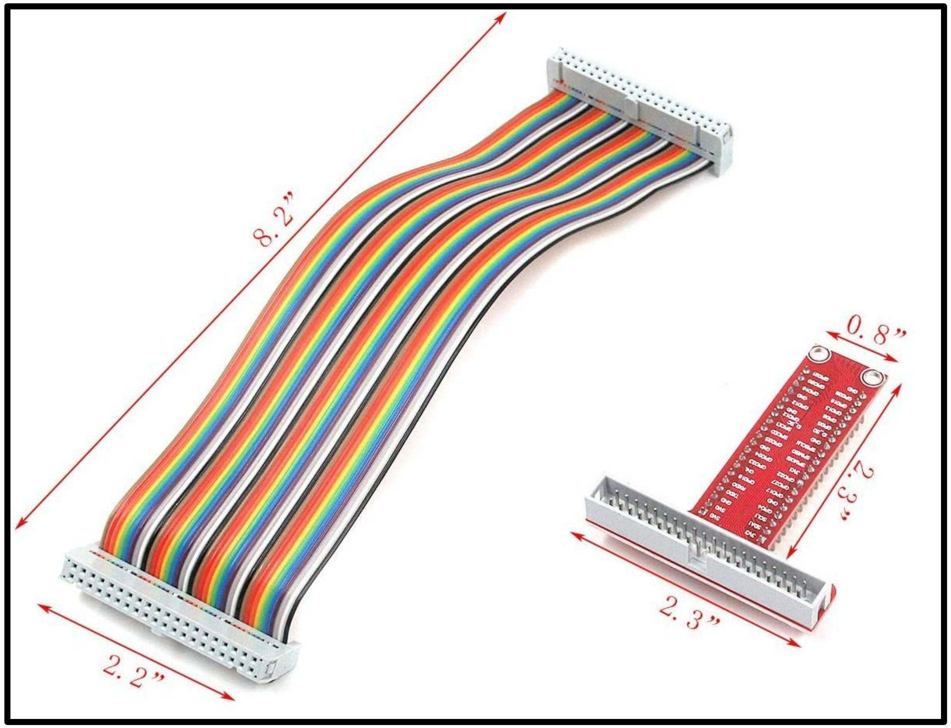
- Camera connector on Jetson Orin Nano: 22 pins
- PI Cam—2: 15 pin connector
- Need 15 to 22 pin cable

Jetson Orin Nano: 40-Pin Expansion Header



3.3V	1	2	5.0V
I2C1_SDA	3	4	5.0V
I2C1_SCL	5	6	GND
GPIO09	7	8	UART1_TXD
GND	9	10	UART1_RXD
UART1_RTS*	11	12	I2S0_SCLK
SPI1_SCK	13	14	GND
GPIO12	15	16	SPI1_CS1*
3.3V	17	18	SPI1_CS0*
SPI0_MOSI	19	20	GND
SPI0_MISO	21	22	SPI1_MISO
SPI0_SCK	23	24	SPI0_CS0*
GND	25	26	SPI0_CS1*
I2C0_SDA	27	28	I2C0_SCL
GPIO01	29	30	GND
GPIO11	31	32	GPIO07
GPIO13	33	34	GND
I2S0_FS	35	36	UART1_CTS*
SPI1_MOSI	37	38	I2S0_DIN
GND	39	40	I2S0_DOUT

GPIO Breakout Expansion Board



GPIO Breakout Expansion Board, Ribbon Cable, T Adapter

Jetson Orin Nano: Tech. Specs.

	Jetson Orin Nano 8GB Module
AI Performance	67 TOPS
GPU	NVIDIA Ampere architecture with 1024 CUDA cores and 32 tensor cores
CPU	6-core Arm® Cortex®-A78AE v8.2 64-bit CPU 1.5MB L2 + 4MB L3
Memory	8GB 128-bit LPDDR5 102GB/s
Storage	Supports SD card slot and external NVMe
Video Encode	1080p30 supported by 1-2 CPU cores
Video Decode	1x 4K60 (H.265) 2x 4K30 (H.265) 5x 1080p60 (H.265) 11x 1080p30 (H.265)
Power	7W–25W

	Reference Carrier Board
Camera	2x MIPI CSI-2 22-pin camera connectors
PCIe	M.2 Key M slot with x4 PCIe Gen3 M.2 Key M slot with x2 PCIe Gen3 M.2 Key E slot
USB	USB Type-A connector: 4x USB 3.2 Gen2 USB Type-C connector for UFP
Networking	1xGbE connector
Display	1x DP 1.2 (+MST) connector
Other I/O	40-pin expansion header (UART, SPI, I2S, I2C, GPIO) 12-pin button header 4-pin fan header DC power jack
Mechanical	103mm x 90.5mm x 34.77mm (Height includes feet, carrier board, module, and thermal solution)

Jetson Orin Nano: Libraries and APIs

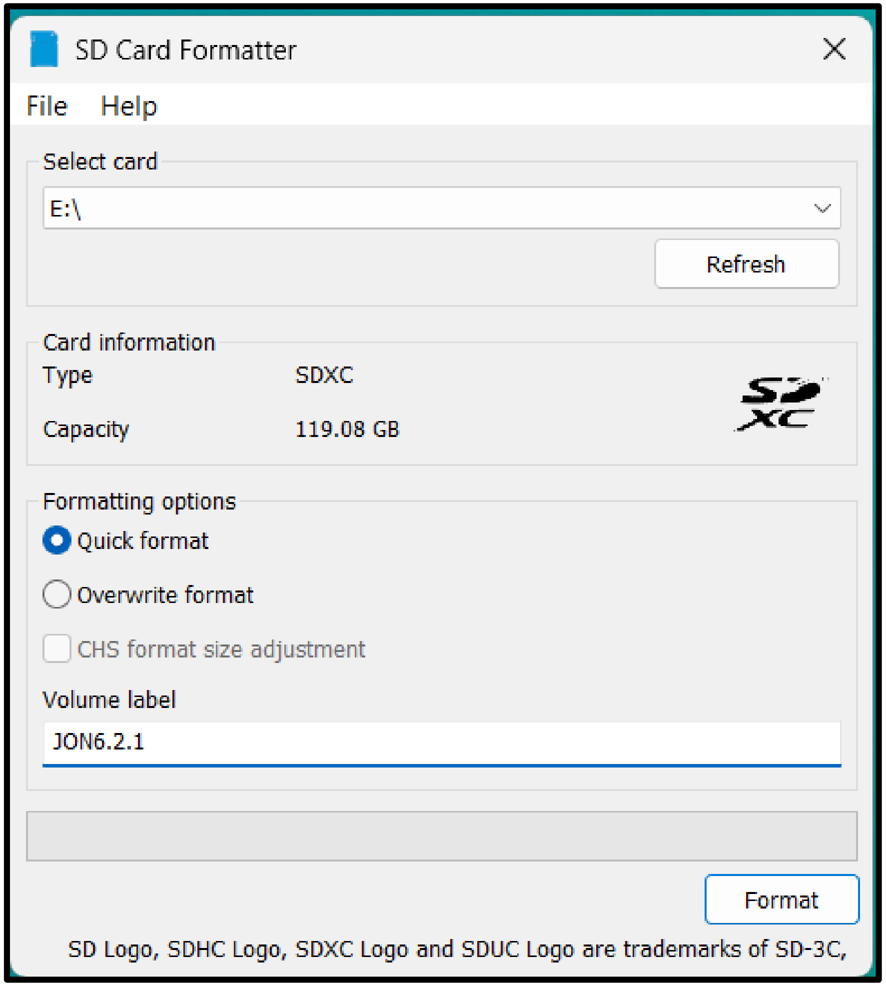
- **AI and deep learning**
  - CUDA Toolkit for GPU-accelerated computing
  - TensorRT for optimizing deep learning inference
  - cuDNN, a CUDA library, providing primitives for frameworks like TensorFlow and PyTorch
- **Computer vision and image processing**
  - Vision Programming Interface (VPI): Algorithms across hardware engines with C++ and Python APIs
  - OpenCV and VisionWorks SDK: Image processing optimized for GPU
  - VisionWorks SDK: optimized for GPU
  - Argus, a low-level API for camera control
- **Multimedia and graphics API packages**
  - libargus for camera capture
  - V4L2 API for video encoding/decoding
  - GStreamer plugins
- **System and utility API**
  - Jetson API library for GPIO pins
  - NvSci for communication between hardware blocks and APIs

Jetson Orin Nano: Libraries and APIs

Jetpack Component	Sample location on reference filesystem
cuDNN	/usr/src/cudnn_samples_<version>/
TensorRT	/usr/src/tensorrt/samples/
CUDA	/usr/local/cuda-<version>/samples/
Multimedia API	/usr/src/jetson_multimedia_api
OpenCV	/usr/share/opencv4/samples/
VPI	/opt/nvidia/vpi<version>/samples
Graphics Demos	/usr/src/nvidia/graphics_demos/
Tegra Accelerated Libraries	/usr/lib/aarch64-linux-gnu/tegra/
Standard SDK Libraries	/usr/lib/aarch64-linux-gnu/

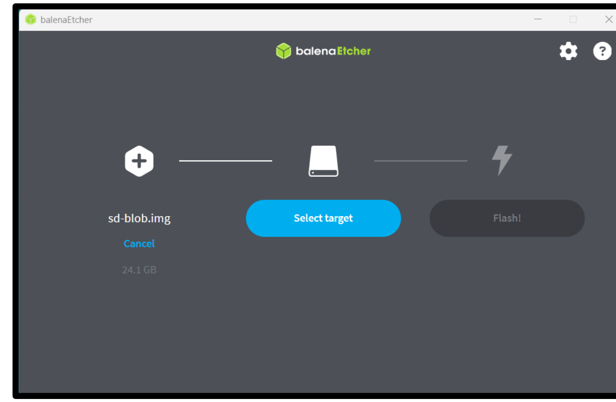
Jetson Orin Nano Super: Flashing Jetpack

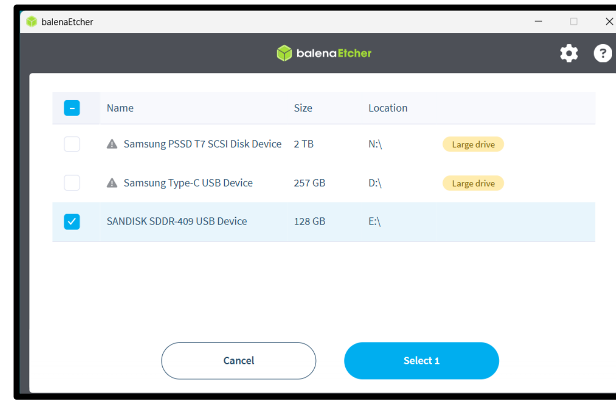
- **Format microSD card using SD Memory Card Formatter**
  - **Download, install, and launch SD [Memory Card Formatter for Windows]**
  - **Select card drive**
  - **Select "Quick format"**
  - **Leave "Volume label" blank**
  - **Click "Format" to start formatting, and "Yes" on the warning dialog**

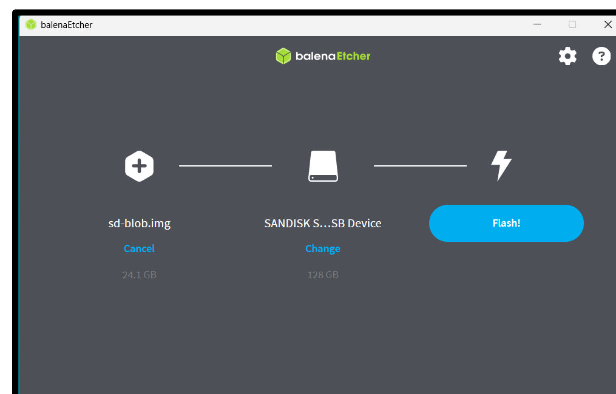


Jetson Orin Nano Super: Flashing Jetpack

- Use Etcher to write the SD card image to your microSD card
  - Leave your microSD card in your Windows computer from the step in previous slide
  - Download, install, and launch Balena Etcher
  - Download Jetpack 6.2.1
  - Click Select image and choose the zipped image file of Jetpack 6.2.1
  - Click “Select drive” and choose the drive corresponding to SD card
  - Click “Flash!” to write and validate the image on SD card







See https://developer.nvidia.com/embedded/learn/get-started-jetson-orin-nano-devkit#write

Jetson Orin Nano Super: Flashing Jetpack

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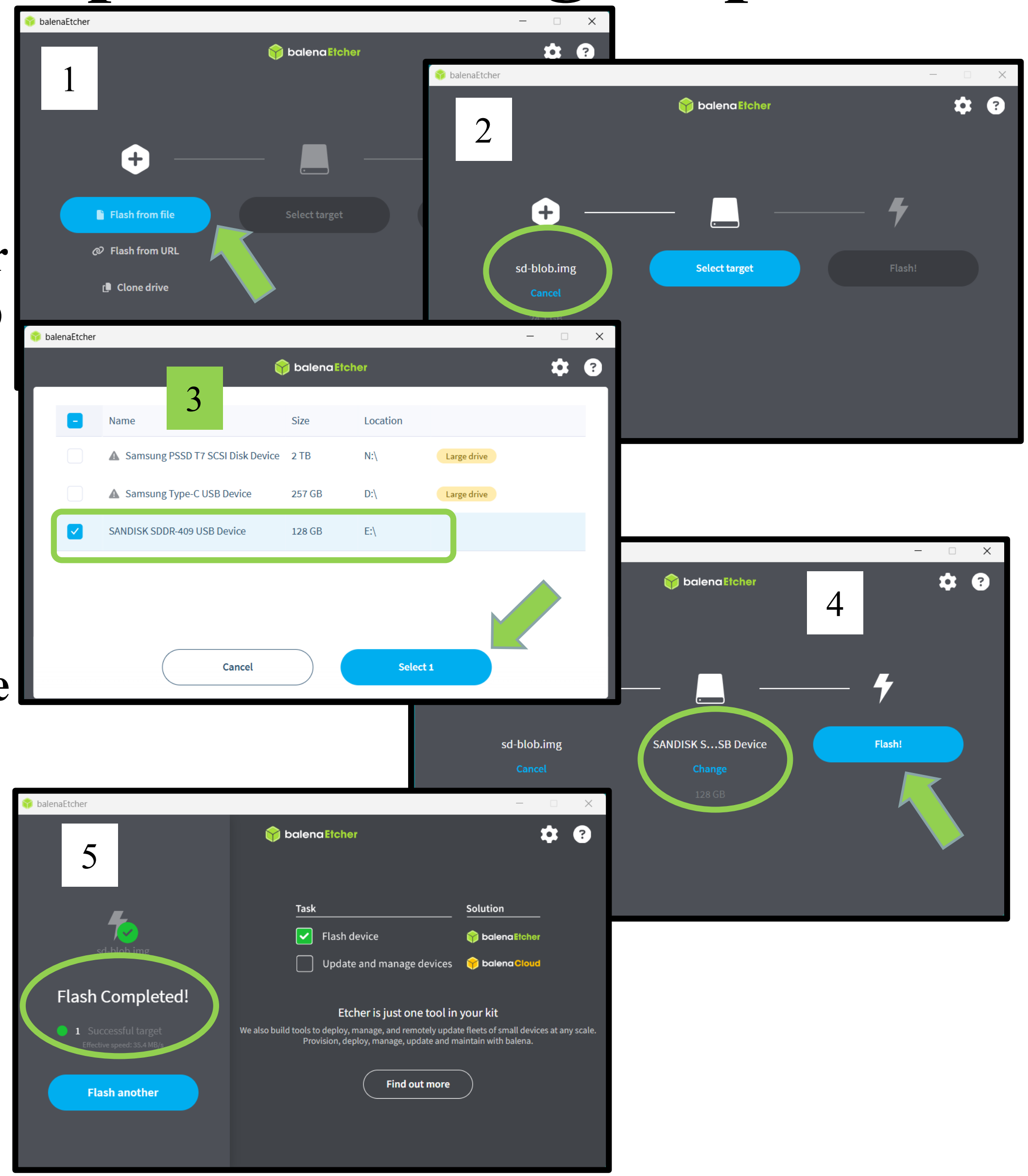
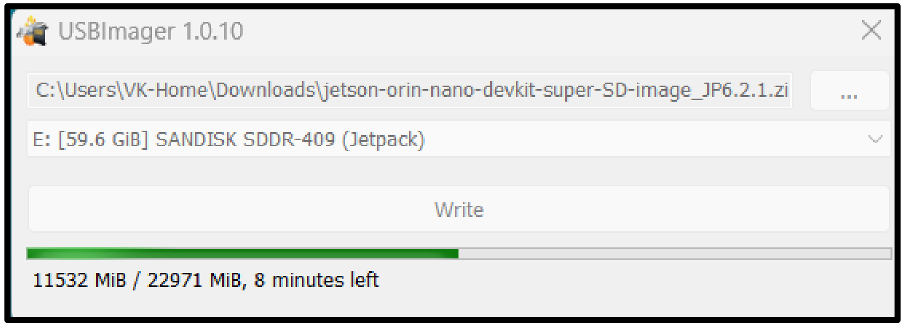


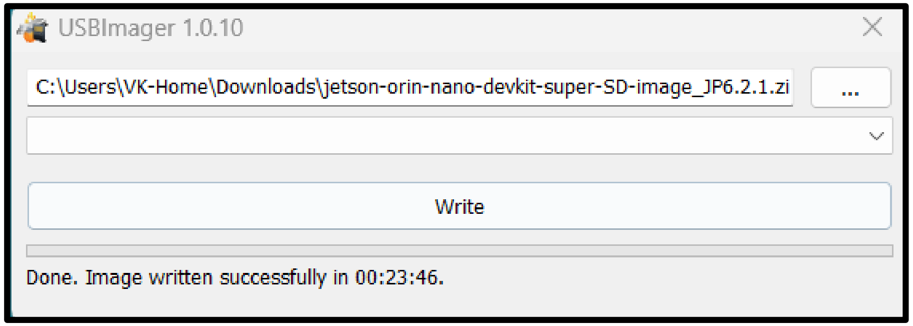
See

https://developer.nvidia.com/embedded/learn/get-started-jetson-orin-nano-devkit#write

Jetson Orin Nano Super: Flashing Jetpack

- Use USBImager to write the SD card image to your microSD card
  - Leave your microSD card in your Windows computer from the prior steps
  - Download and launch [USBImager]
  - Download [Jetpack 6.2.1]
  - In the top box of USBImager, select the zipped image file of Jetpack 6.2.1
  - In the middle box, choose the drive corresponding to your SD card
  - Click “Write” to write the image on SD card

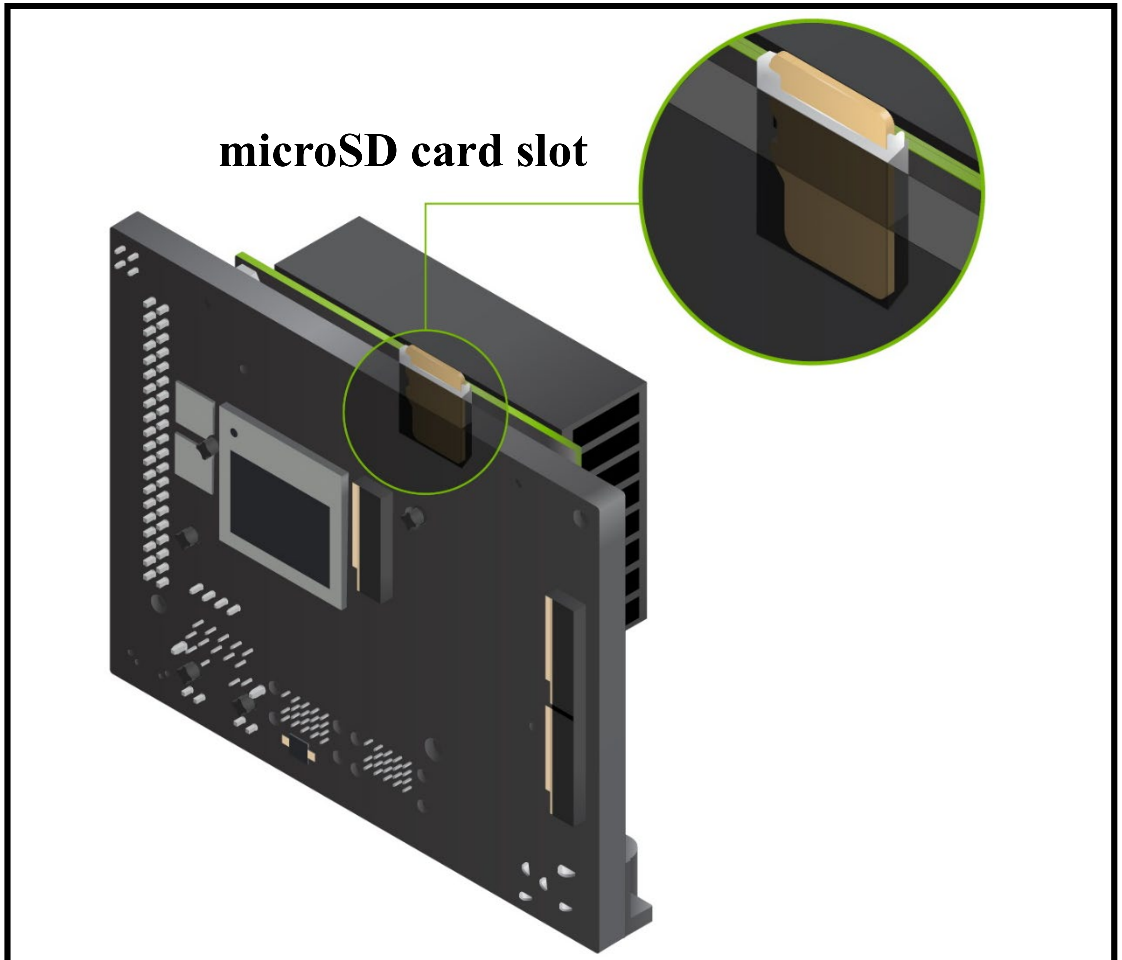




See: https://bztsrc.gitlab.io/usbimager/

Install SD Card

- **Insert the microSD card, with the Jetpack 6.2 flashed, into the microSD card slot of the Jetson Orin Nano module**



Install Monitor, Keyboard/Mouse, Power I/P

- **Connect your computer display using a DP connector to the Jetson Orin Nano**
- **Connect the Bluetooth dongle of your BT keyboard and mouse to a USB port of the Jetson Orin Nano**
- **Connect the Jetson Orin Nano power supply to a power outlet and the Jetson Orin Nano Developer Kit will power on and boot automatically**

Power

DP

BT-USB Dongle

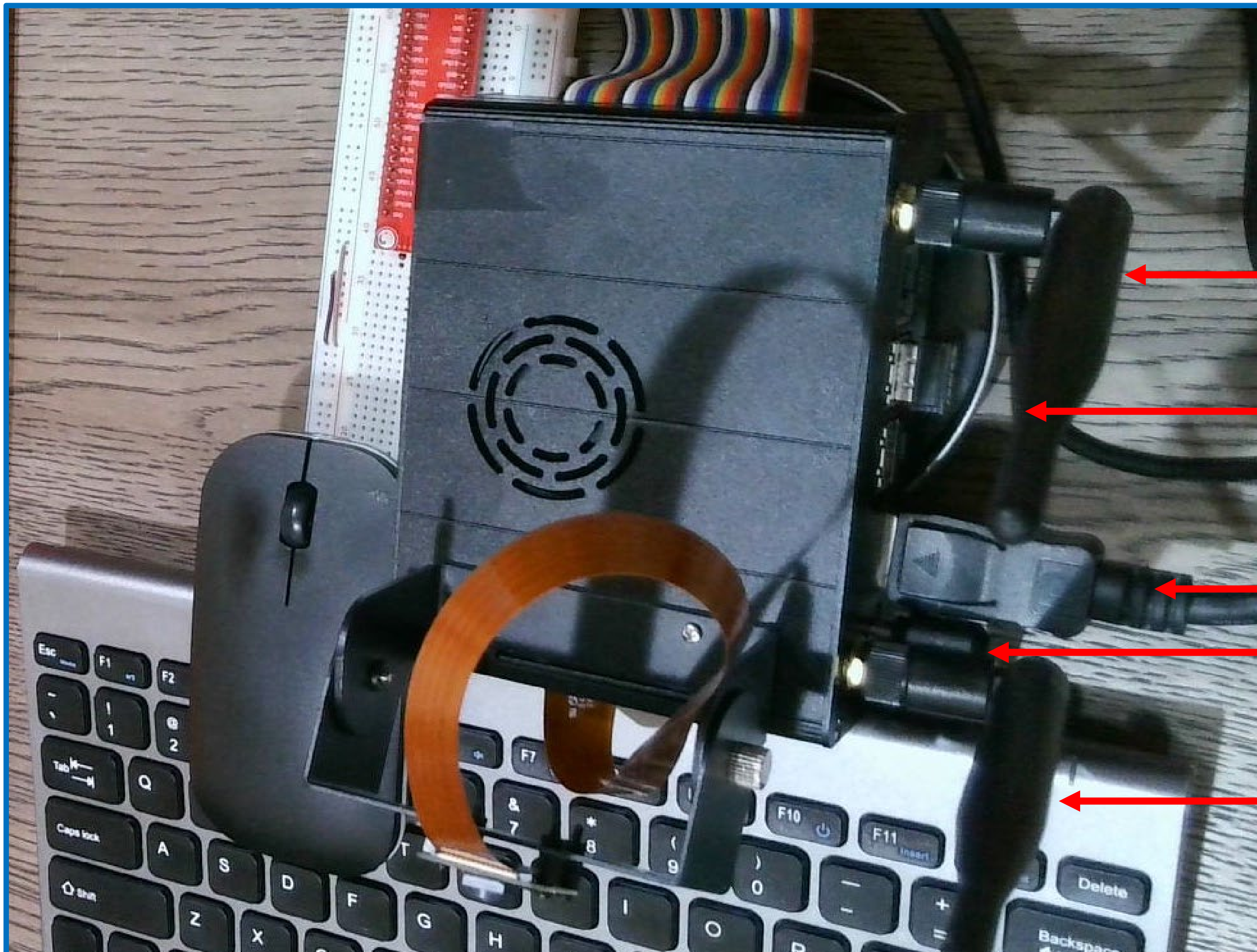
Wi-Fi Antenna

BT Keyboard /Mouse

Wi-Fi Antenna

Wi-Fi Antenna

Install Monitor, Keyboard/Mouse, Power I/P



Wi-Fi Antenna

BT-USB Dongle

DP

Power

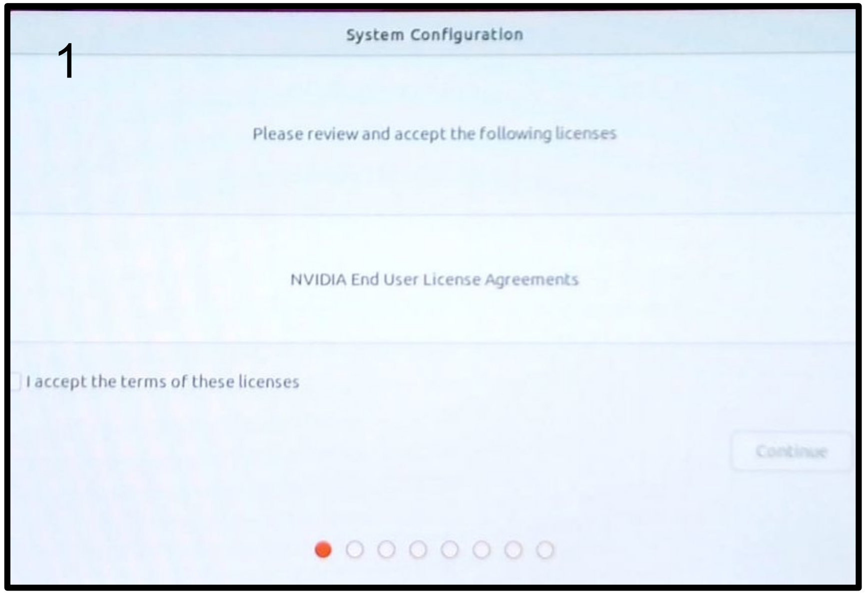
Wi-Fi Antenna

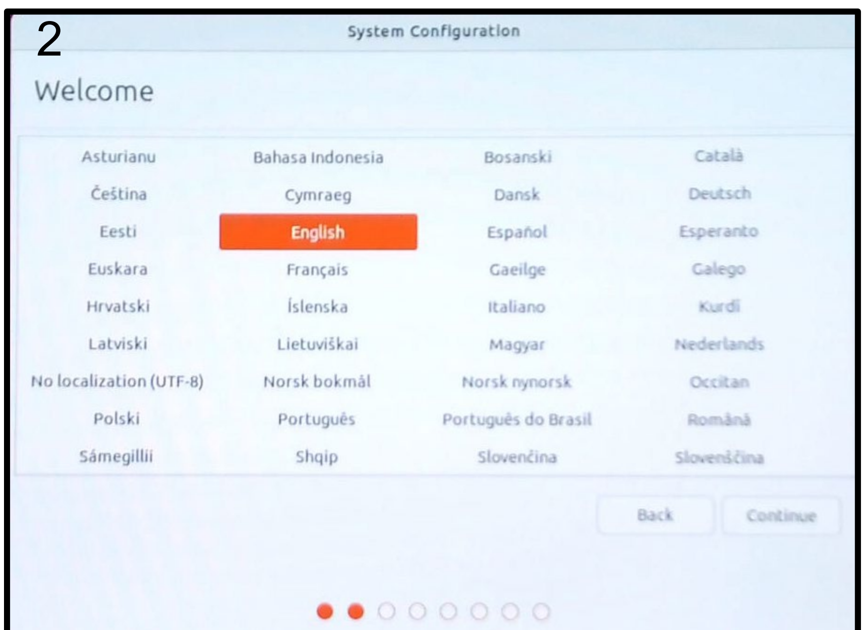
BT Keyboard /Mouse

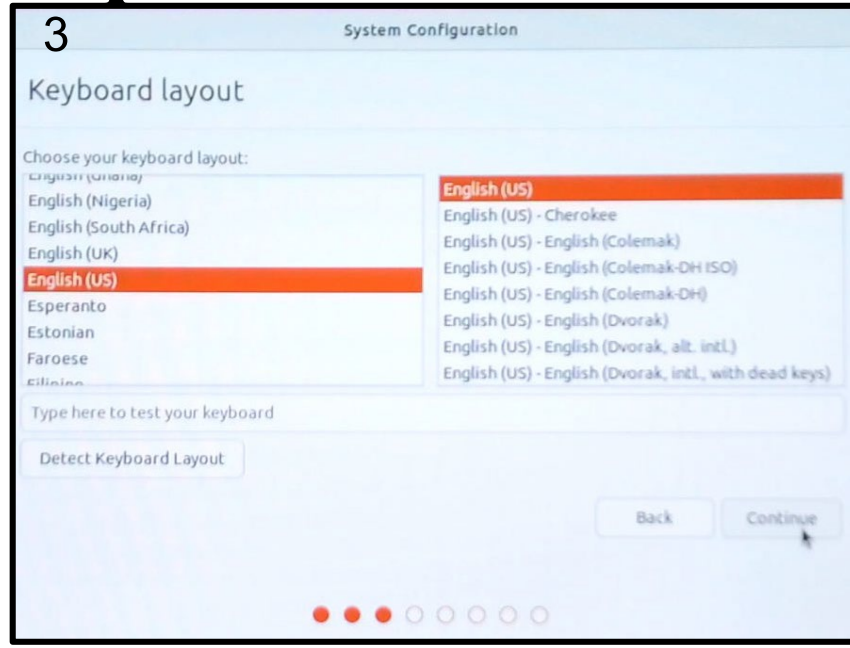
First Boot and Setup—I

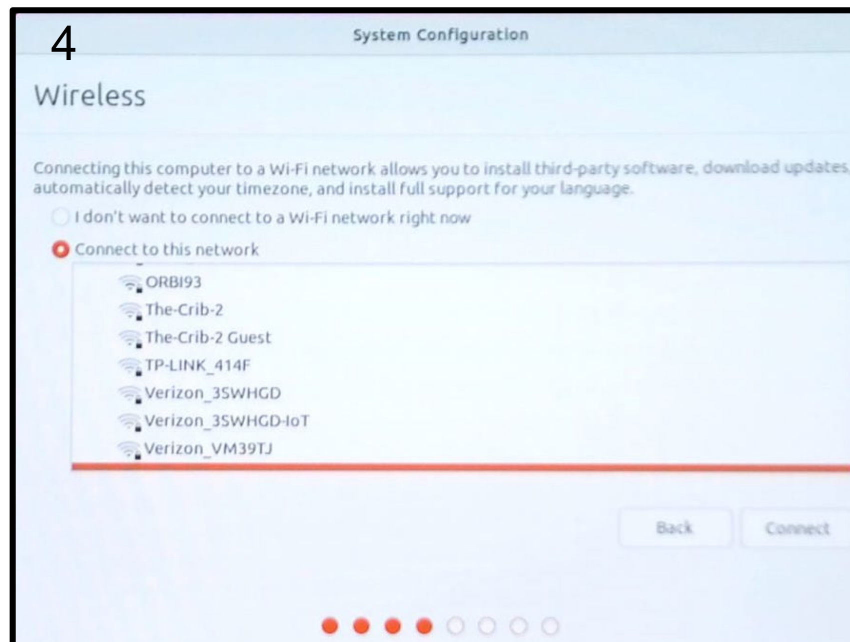
- First bootup with a new SD card launches initial setup
- Review and accept: NVIDIA Jetson software license
- Select system language, keyboard layout, and time zone
- Select from the list of identified Wireless network one that you wish to use
- Setup for the Jetson Orin Nano device your username, password, and computer name
- Select APP partition size for the SD card, default selected size is maximum permissible
- Chromium browser, online accounts, Ubuntu Pro, …
- Restart and log-in

Jetpack Setup—I

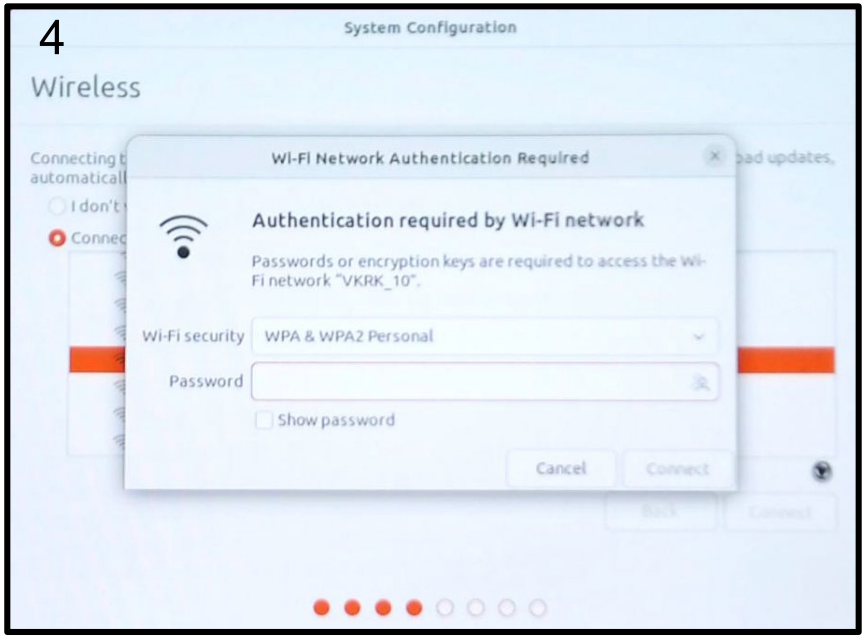


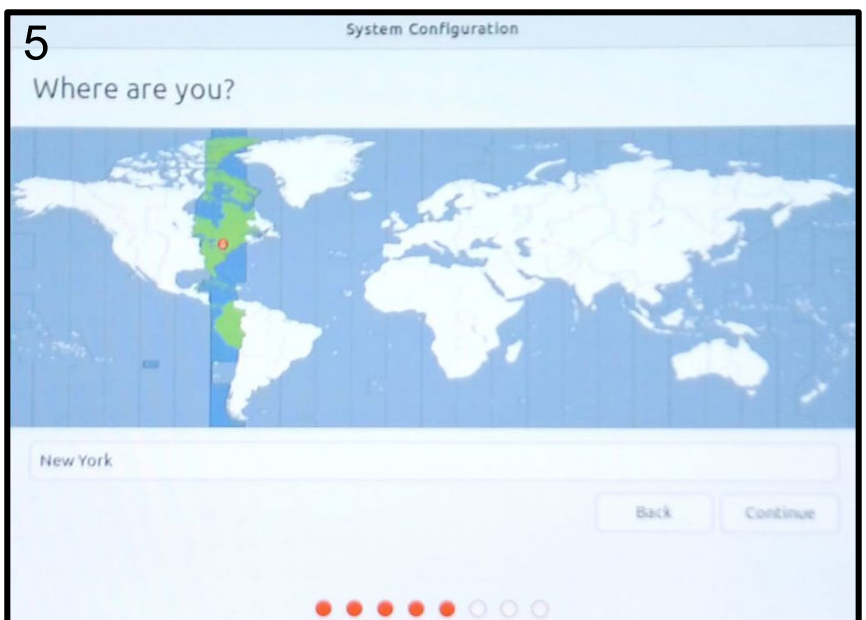


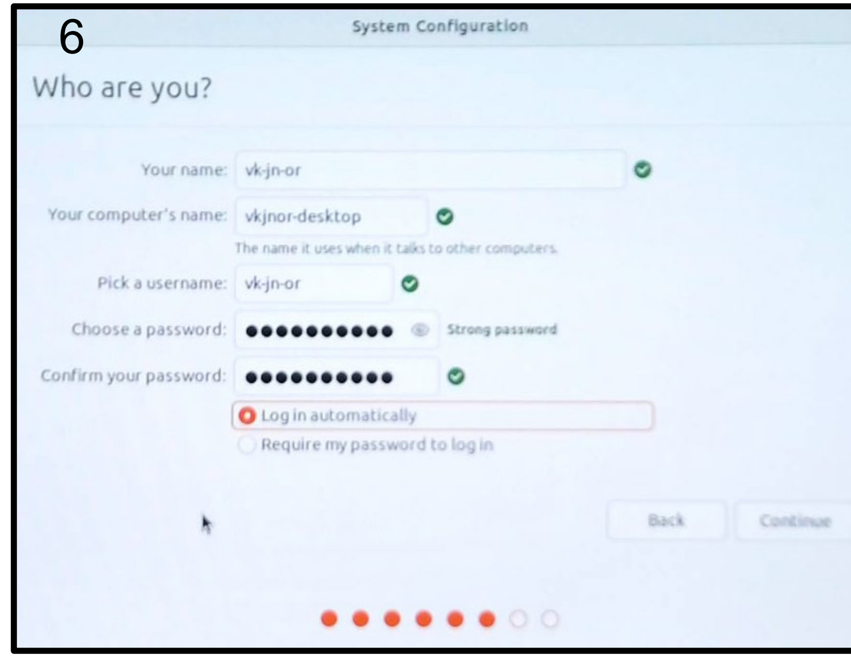


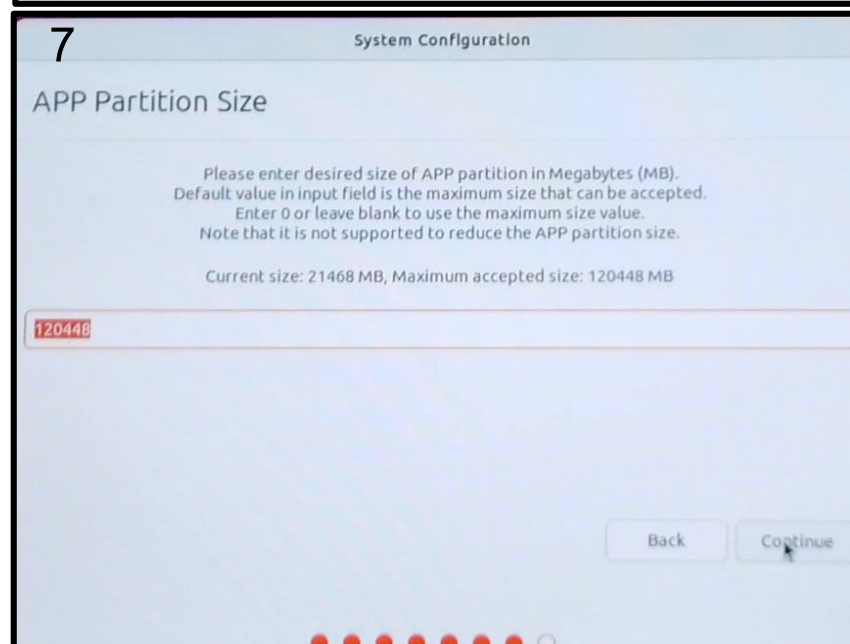


Jetpack Setup—II

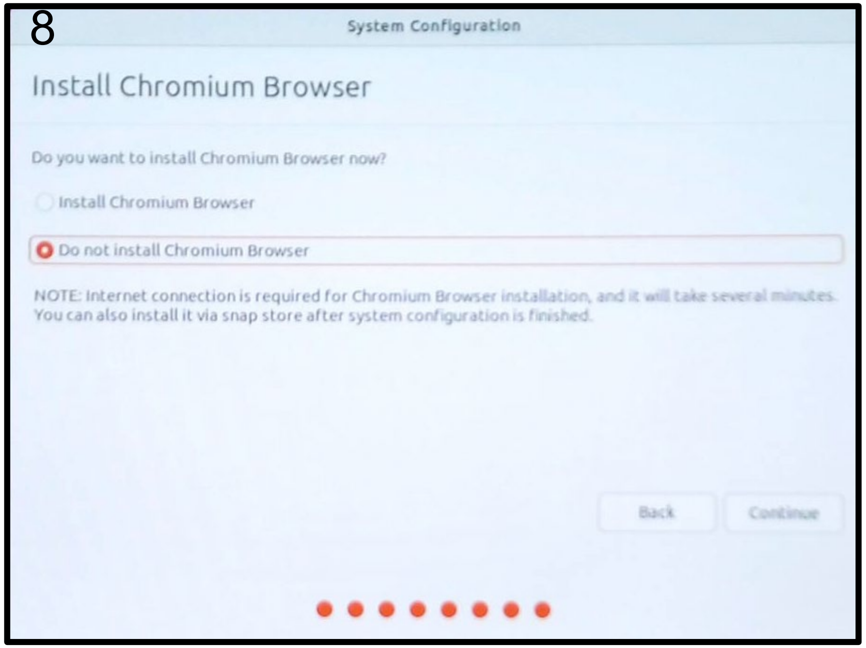


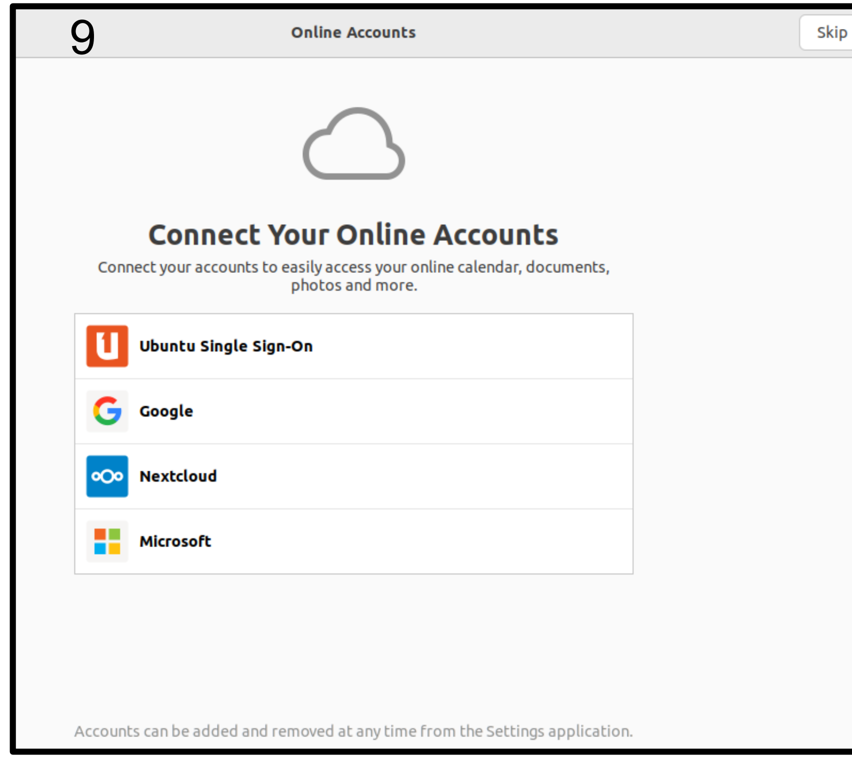






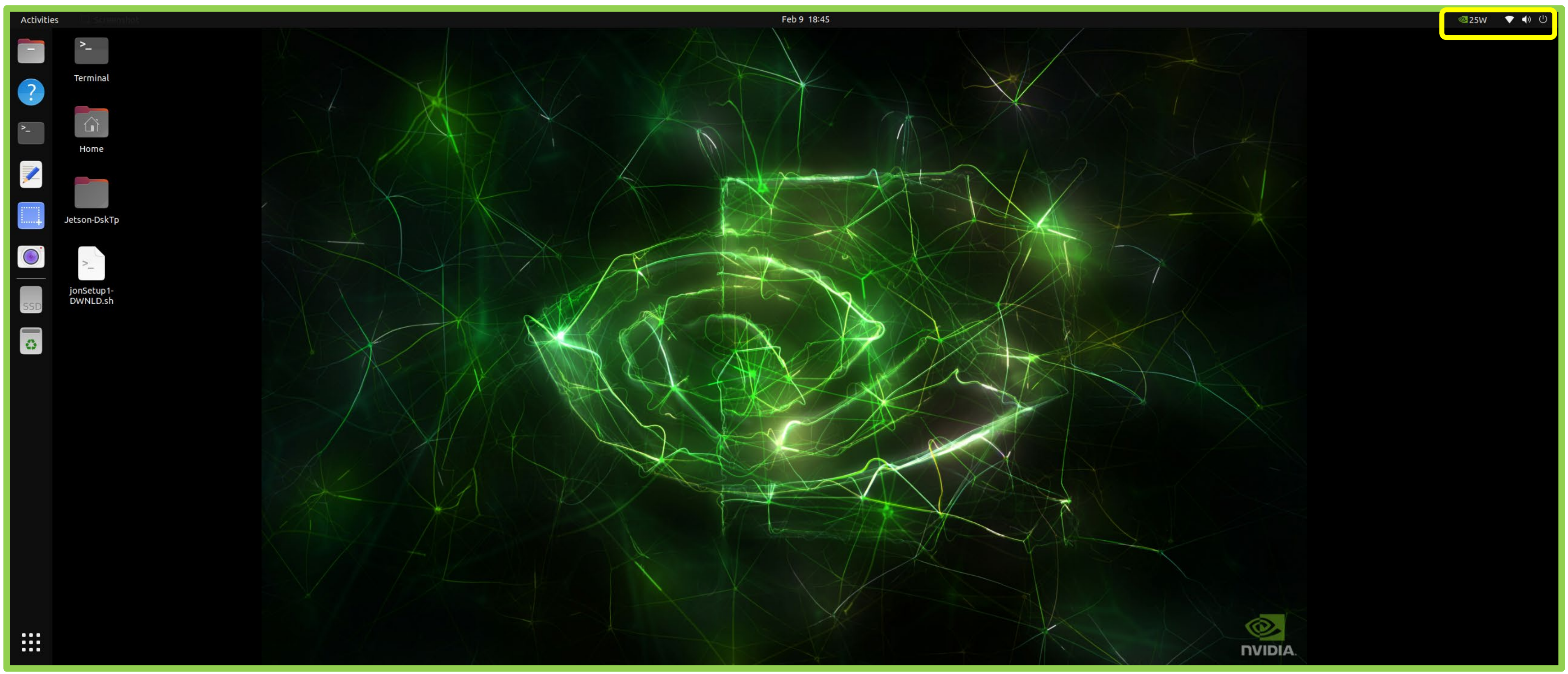
Jetpack Setup—III





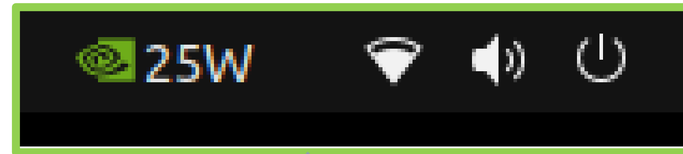
Power Mode Setup—I

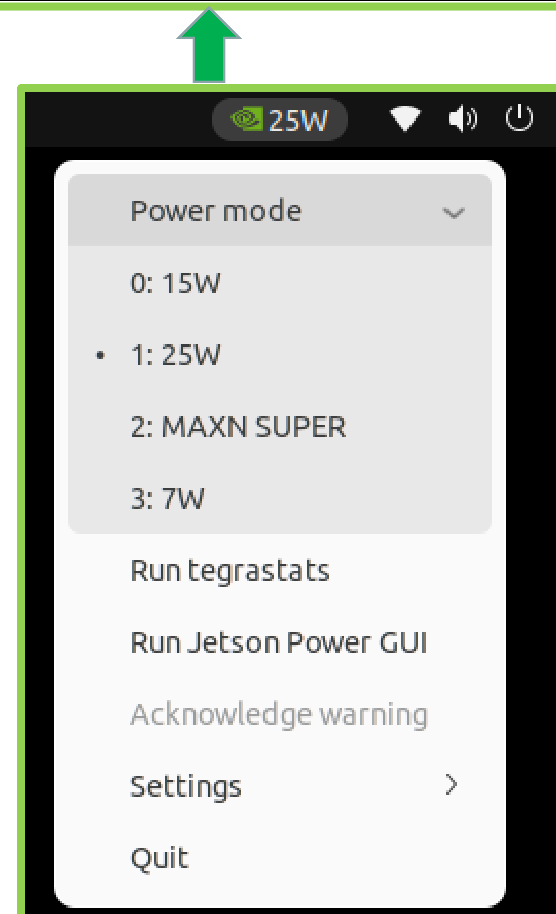
- **Click on the current power mode (25W mode) by clicking the NVIDIA icon on the right side of the Ubuntu desktop's top bar.**



Power Mode Setup—II

- **Select Power Mode from the menu.**
- **Choose MAXN SUPER to enable maximum performance**





Jetson Orin Nano: Super Performance